SWISS ORACLE USER GROUP

SOUG www.soug.ch

Newsletter 5/2014

Sonderausgabe

- OBIF
- DB licensing with VMware
- Delphix
- 12c:
  SQL Plan / Security Features

Pour les alémaniques

Das Schweizer ICT-Magazin für Business-Entscheider
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Für die Romands

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Das Schweizer ICT-Magazin für Business-Entscheider

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Gaetano Bisaz, SOUG

Editorial

Juhui – alles anders, alles neu im 2015! – Juhui

Liebe Leserin, lieber Leser

Nachdem Oracle mit 12c und weiteren neuen Versionen ordentlich vorgelegt hat, ist es an der Zeit, dass die SOUG auch nachzieht und mit neuen Konzepten und Ideen den Kontakt zu Ihnen erstellt. So freut es mich zu berichten, dass der Newsletter 2015 im neuen Gewand erscheint. Deshalb ist dies auch die Ausgabe 5/2014. Wir arbeiten enger mit der DOAG zusammen – unsere Kollegen in Deutschland freuen sich riesig auf die Fachartikel aus der Schweiz, genau wie wir uns über ein grösseres Publikum für unsere Autoren freuen. Die Sprachvielfalt wird erhalten durch intensivere Einbindung der elektronischen Medien.

Es freut mich, dass in dieser Ausgabe ein wahrlich grosser Artikel über BI drin ist. Alexandre Tacchini, Benjamin Gaillard, Fabien Grange haben ein wahres «Werk» abgeliefert. Themaerweiterung, viel Arbeit und gutes Timing = Best Newsletter Article Award. Prost!

Feedback und Ihre Berichte bitte wie immer an nl@soug.ch

Redaktionsschluss und weitere Angaben für den NL2015 demnächst auf unserer Website, XING und LinkedIn.

Stay tuned!

Gaetano Bisaz

Yeah – Tout est différent, tout est neuf en 2015!

Chères lectrices, chers lecteurs

Après qu'Oracle ait présenté dignement la 12c ainsi que d'autres nouvelles versions, il est temps que la SOUG suive le mouvement en créant un contact plus étroit avec vous au travers de nouveaux concepts et de nouvelles idées. Je suis ravi de vous annoncer que la newsletter 2015 fera peau neuve. De ce fait, ceci est le numéro 5/2014. Nous travaillerons en plus étroite collaboration avec la DOAG – nos collègues allemands se réjouissent pour les articles spécialisés venant de la Suisse, tout comme nous nous réjouissons d'avoir un public plus large pour nos auteurs. La diversité linguistique est obtenue grâce à une intégration plus intensive des médias électroniques.
Il est réjouissant qu'il y ait dans cette édition un grand article sur la BI. Alexandre Tacchini, Benjamin Gaillard, Fabien Grange ont délivré une vraie «oeuvre». Élargissement du thème, beaucoup de travail et un bon timing = Best Newsletter Article Award. A votre santé!

Merci d'adresser vos feedbacks et rapports comme toujours à nl@soug.ch.

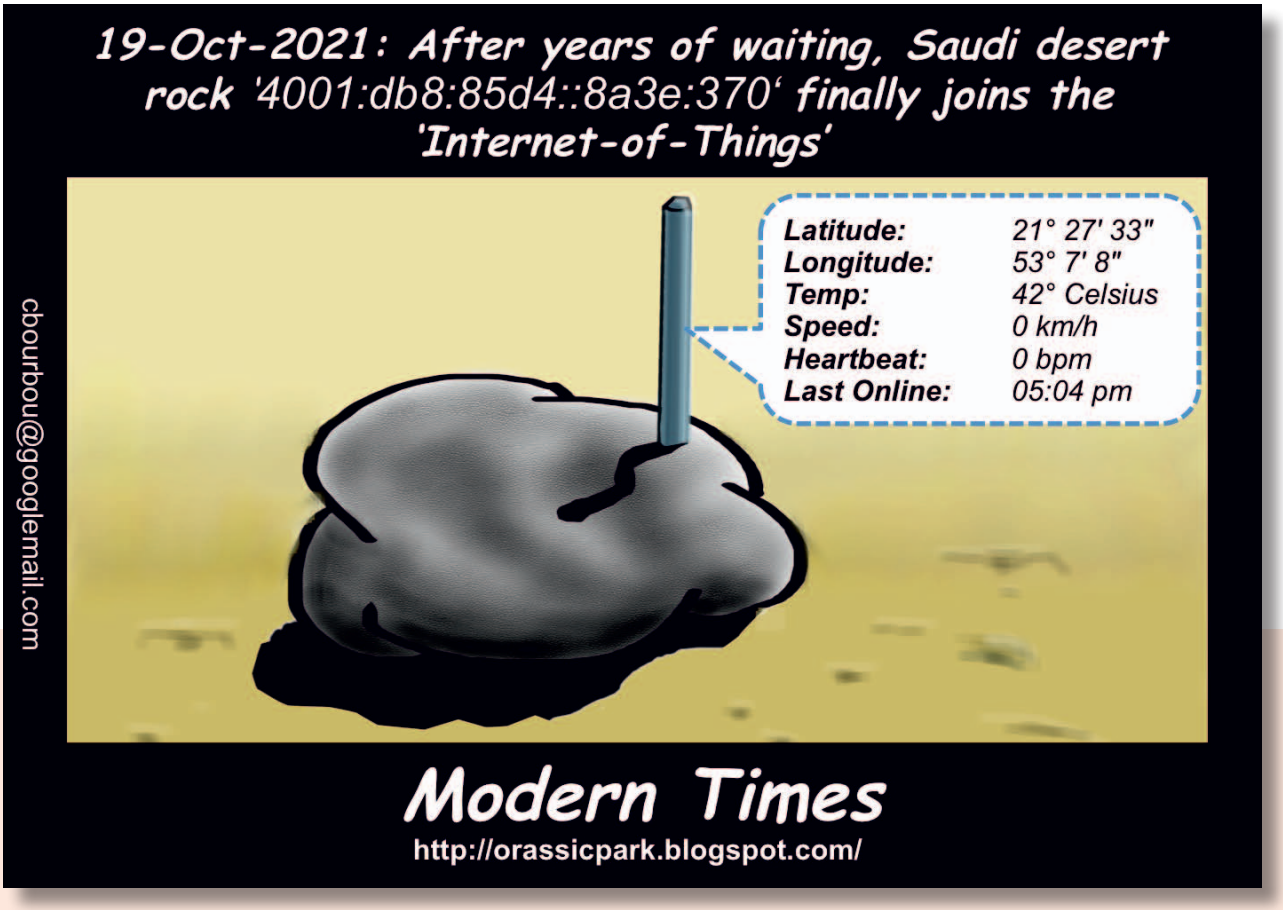
Clôture de la rédaction et plus d'informations pour le NL2015 prochainement sur notre website, XING et LinkedIn.

Restez à l'écoute!

Gaetano Bisaz

Neumitglieder
Nouveaux membres

Oliver Fuhrer	Swisscom
Senad Kaljic	Swisscom
Markus Kauch	Swisscom
Christoph Lutz	Swisscom
Iris Albrecht	Rahn & Bodmer Co.
Roland Matter	Post CH AG
Sabine Heimsath	its-people
Stephan Rudolph	Rudolph AG



Informationen des Vorstands

Liebe SOUG-Mitglieder

Mit dem Jahr 2014 ist für die SOUG ein Jahr mit grossen Veränderungen vergangen. Tony Fräfel hat nach vielen Jahren das Präsidium an seinen Nachfolger Peter Gübeli übergeben und Ansgar Wollnik musste aufgrund seines Wechsels zu Oracle ebenfalls aus dem Vorstand austreten.
Der SOUG-Vorstand hat sich nebst den üblichen Vereinsaktivitäten im laufenden Jahr intensiv mit unseren Stärken und Schwächen beschäftigt. Die wichtigsten Punkte sollen kurz erwähnt sein:

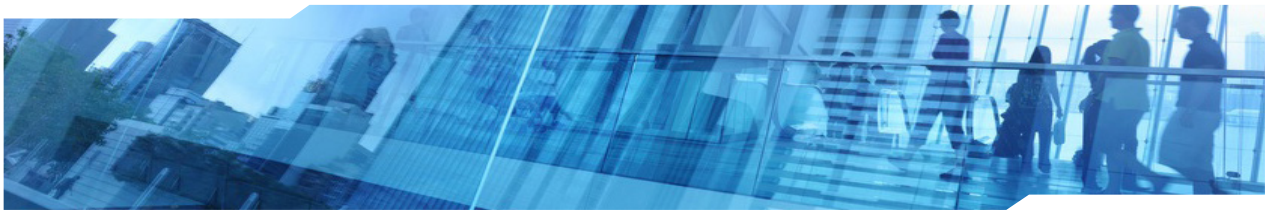
Stärken
- Solide Mitgliederbasis
- Bewährte Konzepte für Fachevents (SIG, SOUG bi dä Lüt, DOAG)
- Regelmässiger Newsletter mit qualitativ gutem Inhalt
- Plattform für Informations- und Erfahrungsaustausch
- Rabatte bei Partnern

Schwächen / Potential
- Nicht komplettes Oracle Themenportfolio abgedeckt
- Bekanntheitsgrad der SOUG ungenügend
- Geringe Nutzung neuer Kommunikationsmedien
- Ressourcen aus dem Vereinsvorstand durch unentgeltliche Arbeiten begrenzt

Um unsere Stärken zu erhalten und gleichzeitig einige Schwächen zu eliminieren, werden wir 2015 verschiedene Aktivitäten anstossen. Ziel ist es, den Mehrwert von SOUG für Sie, unsere über 400 Mitglieder, zu erhöhen und auch die Möglichkeit für ein moderates Wachstum zu schaffen.
Folgende Anpassungen sind deshalb geplant:
- Newsletterzusammenlegung mit der DOAG:
- neu 6x pro Jahr in einer umfangreicheren Ausgabe als heute
- Separate Beilage in französischer Sprache mit Abstracts und Leadartikel
- alle Artikel online in deutscher und Originalsprache verfügbar
- Erweiterung unserer Events für das

ANZEIGE

>>> Account-Manager - Deutsche Schweiz (m/w)



Sind Sie eine strategisch denkende Verkaufspersönlichkeit mit mehrjähriger Erfahrung und Erfolgsausweis im Vertrieb. Kennen Sie sich sehr gut im Schweizer IT-Markt aus?

Für diese anspruchsvolle Funktion benötigen Sie:
> ausgeprägte konzeptionelle Fähigkeiten und unternehmerisches Denken
> eine praxisnahe Verkaufs- und Marketingausbildung
> Erfahrung in der Betreuung von Grosskunden

Im Idealfall sind Sie bereits als Account Manager im Verkauf von Dienstleistungen tätig oder haben ein ausgeprägtes Verständnis der IT-Herausforderungen der Kunden in der Schweiz.

Zu Ihren Hauptaufgaben gehören:
> Auf-, Ausbau und Pflege einer positiven und langjährigen Kundenbeziehung
> Akquirieren von neuen Projekten
> Unterstützung Ihrer Kunden bei Strategie-Entscheiden

Mehr über diese interessante Stelle erfahren Sie unter: www.diso.ch/jobs.

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applikatorische Oracleumfeld (Middleware, Architektur)
- Erweiterung unserer Events für Management- und Strategiethemen rund um Oracle
- Neubesetzung der Abgänge im Vorstand und evtl. Erweiterung, um neue Themen abzudecken
- Noch in Diskussion: Anpassung der SOUG-Homepage auf heutige Bedürfnisse (Mobilfähigkeit, Social-Media Integration, Oraclespezialisten Job-Börse usw.)

Wir hoffen, Ihnen damit einen noch besseren Service zu bieten und sind auf Ihr Feedback gespannt.

Allen ein gutes und erfolgreiches Neues Jahr

Der SOUG Vorstand

Peter Gübeli, Präsident
Thierry Bosshart, Vizepräsident
Gaetano Bisaz, Newsletter
Hans-Jörg Bütler, Events
Yann Neuhaus, Koordination SOUG-R
Kurt Meier, Kassier. ■

Informations du conseil d'administration

Chers membres du SOUG

L'année 2014, riche en événements, touche à sa fin. Après plusieurs années, Tony Fräfel a remis la présidence à son successeur Peter Gübeli. Ansgar Wollnik a dû également quitter le directoire à cause de son nouveau poste chez Oracle.
A part les diverses activités associatives, le directoire du SOUG s'est penché intensément sur les forces et les faiblesses. Voici en bref les points les plus importants:

Forces
- base solide au niveau effectif
- des concepts éprouvés pour les événements métier (SIG, «SOUG chez les gens», DOAG)
- une newsletter régulière avec un contenu de qualité
- une plateforme pour l'échange d'information et d'expériences
- des rabais pour les partenaires

Faiblesses / potentiel
- le portefeuille de thèmes Oracle n'est pas totalement couvert
- le degré de notoriété du SOUG est encore insuffisant
- les nouveaux médias de communication sont faiblement utilisés
- les ressources du directoire, basées sur le bénévolat, sont limitées

Afin de maintenir nos forces et d'éliminer certaines faiblesses, nous allons introduire diverses activités en 2015. L'objectif est d'augmenter la plus-value du SOUG pour vous, nos plus de 400 membres, et également de créer les conditions d'une croissance modérée.
Les ajustements suivants sont ainsi planifiés:
- fusion des newsletters avec le DOAG:
- dorénavant 6 fois par an avec une édition plus volumineuse qu'actuellement
- encart en langue française avec résumés et éditorial
- tous les articles seront disponibles en ligne en version originale et allemande
- extension de nos événements vers le domaine des applications Oracle (middleware, architecture)
- extension de nos événements vers des thèmes stratégiques et de management du domaine Oracle
- nouvelle attribution des départs au sein du directoire, voire son extension, afin de couvrir des thématiques nouvelles
- en discussion: Ajustement du site web du SOUG aux besoins d'aujourd'hui (version mobile, intégration des réseaux sociaux, bourse aux emplois pour spécialistes Oracle etc.)

Nous espérons vous offrir ainsi un meilleur service et attendons avec curiosité vos réactions.

Nous vous souhaitons à toutes et à tous une bonne et prospère nouvelle année.

Le directoire du SOUG

Peter Gübeli, Président
Thierry Bosshart, Vice-président
Gaetano Bisaz, Newsletter
Hans-Jörg Bütler, Evénements
Yann Neuhaus, Coordination SOUG-R
Kurt Meier, Trésorier ■

DOAG Konferenz und Ausstellung 2014

Alle Rekorde gebrochen! So könnte man die diesjährige Konferenz der deutschsprachigen Oracle Community von Mitte November in Nürnberg zusammenfassen. Über 2000 Besucher waren an den drei Konferenztagen im NCC und hatten zwischen den 450 Vorträgen die Qual der Wahl. Die neun Themenbereiche BPM, DWH/BI, Datenbank, Development & Tools, Middleware, MySQL, Oracle & SAP, Oracle Infrastruktur sowie Strategie & Business Practices boten jedermann verschiedenster Tätigkeit und Hirarchiestufe viel Praxiswissen und Neuigkeiten aus dem riesigen Oracle-Umfeld. Bei der Programmgestaltung waren nebst der DOAG wiederholt auch die SOUG und AOUG aktiv beteiligt, und auch die Anzahl der Referenten aus der Schweiz war wieder auf einem erfreulich hohen Niveau.

Wem die Vorträge, Keynotes, spontane Unkonferenzen oder Expertenpanel noch nicht genug boten, besuchte die Ausstellung mit 60 Partnern zu Produkten und Dienstleistungen.

Daneben kam auch das Kulinarische und Kulturelle nicht zu kurz: italienischer oder amerikanischer Fastfood, fränkisches oder mediterranes Buffet sowie die (leider schon zu gut bekannte) Mercator-Lounge mit dem exzellenten Kaffee liessen einen dann doch auch mal eine Session verpassen.

Das feine Schöggeli zum Kaffee konnte man sich dann am SOUG-Stand bei einem kleinen Schwatz mit dem SOUG-Vorstand und Sekretariat gönnen.

Auch der Schweizer Abend am Dienstag gehört zur festen Tradition der Konferenz, und auch hier wurde eine Rekordbeteiligung erreicht. Der Bus, welcher uns in die Altstadt von Nürnberg fahren sollte, war bis auf den letzten Platz belegt, sodass die restlichen Teilnehmer noch mit einem PW den Weg zur Marktfrau und Lebkuchenfrau in Angriff nahmen. Leider war uns das Wetter nicht so gut gesinnt, und bei leichtem Regen wurden die zwei Gruppen dann durch die Altstadt geführt und hörten viel Spannendes und Unterhaltsames zur Lebkuchenstadt.

Beim anschliessenden Nachtessen konnten sich alle mehr als genug verköstigen – der Wirt erwartete wohl eine Truppe Schwerarbeiter (siehe Fotos). Interessante Gespräche liessen die Zeit wieder wie im Fluge vergehen und einige sollen wohl die erste Session am Mittwoch «verpasst» haben. An dieser Stelle den Sponsoren diso, dbi services und Oracle nochmals herzlichen Dank für ihre Unterstützung!

Auf Wiedersehen Nürnberg – die Konferenz 2015 findet vom 17.–19.11.15 mit Schulungstag am 20.11.15 statt.

#DOAG2014 vielen Dank für alles! Es war super!

Wiedereingliederung am Arbeitsplatz nach so einer tollen #DOAG2014 fällt heute extrem schwer.

3 Supertage auf der #DOAG2014, viele Top-Vorträge, inspirierende Kundengespräche. Und Lego-Spass bei uns am Stand.

Germany knows how to put on a show. Maybe the best tech conference experience I've had in the last 5 years. #DOAG2014. ■

Arnold Keller, Oracle Software (Schweiz) GmbH

Current Oracle Database Release Schedule

Platform	12.2.0.1	12.1.0.2[11,12]	12.1.0.12	11.2.0.4[10]	11.2.0.3	11.2.0.2	11.2.0.12	11.1.0.7[1]	10.2.0.5[3]	10.2.0.4[4]	10.1.0.5
Linux x86	*Not planned*	*Not planned*	*Not planned*	28. Aug 13	23. Sep 11	13. Sep 10	01. Sep 09	18. Sep 08	30. Apr 10	22. Feb 08	30. Jan 06
Linux x86-64	1HCY2016	22. July 14	25. June 13	27. Aug 13	23. Sep 11	13. Sep 10	01. Sep 09	18. Sep 08	30. Apr 10	17. Mar 08	24. Feb 06
Oracle Solaris SPARC (64-bit)	1HCY2016	22. July 14	25. June 13	29. Aug 13	01. Oct 11	24. Sep 10	06. Nov 09	06. Oct 08	19. May 10	30. Apr 08	05. Feb 06
Oracle Solaris x86-64 (64-bit)	1HCY2016	22. July 14	25. June 13	29. Aug 13	01. Oct 11	24. Sep 10	25. Nov 09	*Not planned*	19. May 10	13. Nov 08	*Not planned*
Microsoft Windows x64 (64-bit)	TBD	25. Sep 14	09. July 13	25. Oct 13	11. Nov 11	15. Dec 10	02. Apr 10	13. Nov 08	27. July 10	16. May 08	*Not planned*
HP-UX Itanium [9]	TBD	14. Nov 14	09. Jan 14	10. Oct 13	29. Oct 11	19. Oct 10	22. Dez 09	06. Oct 08	03. June 10	30. Apr 08	07. June 06
HP-UX PA-RISC (64-bit) [8]	Platform desupported [8]	Platform desupported [8]	Platform desupported [8]	02. Jan 14	16. Feb 12	15. Mar 11	20. May 10	11. Nov 08	15. Dec 10	02. June 08	05. Feb 06
IBM AIX on POWER Systems	TBD	14. Nov 14	09. Jan 14	10. Oct 13	29. Oct 11	19. Oct 10	22. Dez 09	06. Oct 08	03. June 10	15. May 08	05. Feb 06
IBM Linux on System z	TBD	14. Nov 14	09. Jan 14	09. Jan 14	01. Dec 11	30. Mar 11	*Not planned*	*Not planned*	03. Jan 11	16. Dec 08	26. Aug 06
Microsoft Windows (32-bit)	*Not planned*	*Not planned*	*Not planned*	25. Oct 13	11. Nov 11	15. Dec 10	05. Apr 10	10. Oct 08	19. July 10	17. Mar 08	13. Feb 06
Instant Client Releases											
Apple Mac OS X (Intel)	TBD	*Planned*	*Not planned*	20. Apr 14	31. Jan 13 *Instant Client only*	*Not planned*	*Not planned*	*Not planned*	*Sched TBA*	10. Apr 09 *Single Instance only*	*Not planned*
IBM Linux on POWER				04. Dez 14							
Older Releases											
Apple Mac OS X (PowerPC)	*Platform desupported*										08. Jan 07
HP OpenVMS Alpha	*Platform desupported*								31. Oct 12	15. Dec 08	15. Feb 08
HP OpenVMS Itanium	*Platform desupported (see Doc ID 1307745.1)*								31. Oct 12	15. Dec 08	*Not planned*
HP Tru64 UNIX	*Platform desupported*								21. Apr 11	20. Feb 09	18. Oct 06
IBM Linux on POWER	*Platform desupported (see Doc ID 1310584.1)*								17. Mar 11	09. Jan 09	*Not planned*
IBM z/OS on System z	*Platform desupported (see Doc ID 461234.1)*								26. Oct 12	*Not planned*	06. Mar 06
Linux Itanium [9]	*Platform desupported (see Doc ID 1130325.1)*								17. Mar 11	24. Sep 08	30. Apr 06
Microsoft Windows Itanium (64-bit) [9]	*Platform desupported (see Doc ID 1307745.1)*								12. May 11	02. Feb 09	30. Jan 06
Oracle Solaris x86 (32-bit)	*Platform desupported*									14. Nov 08 *Last patch set for this platform*	18. June 06

Notes:

[1] 11.1.0.7 is the last patch set for Release 11.1. Future integrated error correction will come in the form of Patch Set Updates. See Patch Set Updates for Oracle Products (Doc ID 854428.1) for current details.

[2] **Base release** – download from otn.oracle.com.

[3] 10.2.0.5 is the last patch set for Release 10.2. Once patching ends for 10.2.0.4, future integrated error correction will come in the form of Patch Set Updates. See Patch Set Updates for Oracle Products (Doc ID 854428.1) for current details.

[4] The **start of Extended Support** no longer forces customers to run the last patch set in order to receive new fixes (such as PSU or CPU) - 10.2.0.4 will be patched up until the end date listed above.

[5] Patching end date for 10.2.0.4 has been pushed back from the original 30-Apr-2011 date. See the announcement in Database Patch Set 10.2.0.4 patching extended to July 31, 2011 (Doc ID 1293180.1)

[6] **TBD**: Patching end date will be set once the next patch set is released.

[7] Patching end dates for terminal patchsets of a release are equivalent to the end of Extended Support. An Extended Support contract is required to receive patches once a release enters the Extended Support phase. See the Lifetime Support Policy covering Technology Products for dates of Extended Support.

[8] Oracle Database Release 11.2 will be the last release on HP-UX PA-RISC. See Support Status for Oracle Database on HP-UX PA-RISC systems (Doc ID 1313798.1) for details.

[9] Itanium-based systems (HP-UX, Windows and Linux): Additional Extended Support has been made available for these platforms. The patching end date table above reflects the new end of Extended Support. See Oracle Software Technical Support policies for more details.

[10] 11.2.0.4 is the last patch set for Release 11.2. Future integrated error correction will come in the form of Patch Set Updates. See Patch Set Updates for Oracle Products (Doc ID 854428.1) for current details.

[11] 12.1.0.2 is currently release only as Enterprise edition – see doc 1905806.1 for details.

[12] 12.1.0.2 is the last patch set for Release 12.1. Future integrated error correction will come in the form of Patch Set Updates. See Patch Set Updates for Oracle Products (Doc ID 854428.1) for current details.

Future Oracle Database releases are planned for the other platforms but due to Oracle's revenue recognition policy we are unable to publish those schedules at this time. This schedule only includes releases which have already been released on at least one platform.

Last Patch Set: When Oracle knows which patch set will be the last for a particular release, it will be indicated but designation as such is at Oracle's discretion and subject to change without notice.

Solaris x86-32: The last patch set for this platform is 10.2.0.4.

Solaris x86-64: Oracle has elevated the Solaris x86-64 to equal priority with SPARC Solaris for future Database releases. Customers should expect more timely releases and patch sets for the Solaris x86-64 platform

32-bit clients: 32-bit clients are shipped with some patch sets, but the schedule is the same as for the rest of the patch set unless otherwise noted.

Alle Infos sind online verfügbar auf
https://support.oracle.com unter Certifications und in **Note 742060.1 Release Schedule of Current Database Patch Sets**

Costantinos Bourboulas, Oracle Software (Schweiz) GmbH

Availability Development Tools

Oracle Development Tools	Platform	Release	License
JDeveloper	Java, cross platform	12.1.3.0.0	Free
SQL Developer	Java, cross platform	**4.0.3.16.84**	Free
SQL Developer Data Modeler	Java, cross platform	**4.0.3.853**	Free
Enterprise Pack for Eclipse	Java, cross platform	**12.1.3.3**	Free
Developer Tools for Visual Studio	Windows	12.1.0.1.2	Free
BI Publisher	Java, cross platform	11.1.1.7.1	Required
Application Express	All DB Editions >= 10.2.0.4	**4.2.6.00.03**	Free
Forms Builder, Reports Builder	Fusion Middleware	11.1.2.2.0	Required

software downloads: **www.oracle.com > Downloads > Developer Tools**

These statements are for information purposes only **Version:** December 2014

Franck Pachot, dbi services

12c SQL Plan Directives

Sometimes the optimizer does a bad choice because of misestimated cardinalities. 11g has brought the ability to recognize those errors by the mean of cardinality feedback. That was reactive optimization. 12c has introduced a new optimizer component to make this feedback more proactive for future queries. Let me show you SQL Plan Directives lifecycle on a simple example.

Missing statistics

You have collected system statistics that are relevant for your hardware. You have dictionary statistics that are accurate. And you have a job that collects object statistics so that they are always fresh. How can you have missing statistics? Let's see an example.
I create the following table:

```
SQL> create table DEMO_TABLE as select mod(rownum,2) a,mod(rownum,2)
b,mod(rownum,2) c,mod(rownum,2) d from dual connect by level <=1000;

Table created.

SQL>
SQL> select * from DEMO_TABLE where rownum<=5;

         A          B          C          D
---------- ---------- ---------- ----------
         1          1          1          1
         0          0          0          0
         1          1          1          1
         0          0          0          0
         1          1          1          1
```

You see the pattern: all columns are either 0 or 1
I have statistics on them:

```
SQL> select column_name,num_distinct from user_tab_col_statistics where
table_name='DEMO_TABLE';

COLUMN_NAM NUM_DISTINCT
---------- ------------
A                     2
B                     2
C                     2
D                     2
```

And statistics are right: each column has two distinct values. Then let's run the following query:

```
SQL> alter session set statistics_level=all;

Session altered.

SQL> select count(*) count1 from DEMO_TABLE where a=0 and b=0 and c=0
and d=0;

              COUNT1
--------------------
                 500
```

I've 1000 rows. Half of them have all 0 and half of them have all 1.

This is a simple query but I want to check if the optimizer has estimated the cardinalities correctly.

And the execution plan:

```
SQL> select * from table(dbms_xplan.display_cursor(null,null,'iostats
last'));

PLAN_TABLE_OUTPUT
---------------------------------------------------------------------
SQL_ID  86z3n2y70wavc, child number 0
-------------------------------------
select count(*) count1 from DEMO_TABLE where a=0 and b=0 and c=0 and d=0

Plan hash value: 1839825454

--------------------------------------------------------------------
| Id  | Operation          | Name       | Starts | E-Rows | A-Rows |
--------------------------------------------------------------------
|   0 | SELECT STATEMENT   |            |      1 |        |      1 |
|   1 |  SORT AGGREGATE    |            |      1 |      1 |      1 |
|*  2 |   TABLE ACCESS FULL| DEMO_TABLE |      1 |     63 |    500 |
--------------------------------------------------------------------

Predicate Information (identified by operation id):
---------------------------------------------------
   2 - filter(("A"=0 AND "B"=0 AND "C"=0 AND "D"=0))
```

Here you see the problem: the query returns 500 rows but the optimizer estimated them to 63. That can be a problem in a larger query where the number of rows returned will determine the join method.

Where the problem comes from?

- We know that half of the rows (so 500 here) have all values equal to 0
- The optimizer only know about each column

Then, the CBO estimates 'A=0' to return ½ of the rows, then adding 'B=0' returns ½ of the rows etc. So the optimizer estimate the whole combination of 'A=0 and B=0 and C=0 and D=0' to return 1/16th of the rows, which is 63.

Solutions in 11g

What are the solutions for that? In 11g we have two solutions:

- Extended statistics: we calculate statistics on the group of columns (A,B,C,D) so that the optimizer is aware of the correlation
- Dynamic sampling: when dynamic sampling level is at least 4 then complex where clauses like that one will trigger dynamic sampling. And in any sample, the optimizer will see that the predicate returns half of the rows.

12c does not add new solutions for that (except Adaptive Plan that will lower the bad consequence of misestimating) but will bring automation to those two alternatives, in an adaptive way. Let's see Automatic Reoptimization and SQL Plan Directives.

Automatic Reoptimization

If I check the new 'IS_REOPTIMIZABLE' column from V$SQL, I see that it is flagged to 'Y' for that query:

```
SQL> select sql_id,child_number,is_reoptimizable from v$sql where sql_
id='86z3n2y70wavc';

SQL_ID                CHILD_NUMBER I
------------- -------------------- -
86z3n2y70wavc                    0 Y
```

During execution, oracle has seen that the actual number of rows (the A-Rows above) is very different from the estimated ones (E-Rows). Then the statement is flagged for reoptimization: the next execution will hard parse again instead of sharing the cursor which is known to have been built on bad estimations.

And I can have more information with the 'report' format from dbms_xplan:

```
SQL> select * from table(dbms_xplan.display_cursor('86z3n2y70wavc',null,'
report'));
```

I don't reproduce the first part which is the same as the one above, but the 'report' format adds the following:

```
Reoptimized plan:
-----------------
This cursor is marked for automatic reoptimization.  The plan that is
expected to be chosen on the next execution is displayed below.

Plan hash value: 1839825454

-------------------------------------------------
| Id  | Operation          | Name       | Rows  |
-------------------------------------------------
|   0 | SELECT STATEMENT   |            |   1   |
|   1 | SORT AGGREGATE     |            |   1   |
|*  2 | TABLE ACCESS FULL  | DEMO_TABLE | 500   |
-------------------------------------------------

Predicate Information (identified by operation id):
---------------------------------------------------

   2 - filter("A"=0 AND "B"=0 AND "C"=0 AND "D"=0)
```

Not only I know that the statement will be reoptimized, but I also have the future plan.

And you see that the estimations (Rows) are now accurate because the optimizer has used the statistics from the previous execution (this comes from cardinality feedback).

SQL Plan directive

So, the optimizer has seen that there were a cardinality misestimate. That was reactive: the previous execution may have been executed with a bad plan. But that knowledge can help future query executions.

This is what SQL Plan Directives are: it stores persistently this information to help future executions.

Persistent means that it is not only in memory, but stored in the dictionary. So in order to see them, we have to flush them:

```
SQL> exec dbms_spd.flush_sql_plan_directive;

PL/SQL procedure successfully completed.
```

And we can see which information is stored in the dictionary:

```
SQL> select directive_id,type,state,reason,created,last_modified,last_used,notes from dba_sql_plan_directives where
directive_id in(select directive_id from dba_sql_plan_dir_objects where owner='DEMO' );

        DIRECTIVE_ID TYPE             STATE      REASON
-------------------- ---------------- ---------- ----------------------
NOTES
----------------------------------------------------------------------
 1350980939868665098 DYNAMIC_SAMPLING USABLE     SINGLE TABLE CARDINALITY MISESTIMATE
<spd_note>
  <internal_state>NEW</internal_state>
  <redundant>NO</redundant>
  <spd_text>{EC(DEMO.DEMO_TABLE)[A, B, C, D]}</spd_text>
</spd_note>
```

A new directive has been created.

For the moment (12.1) directives are only created for optimizer bad decision for a 'CARDINALITY MISESTIMATE', the type being type 'DYNAMIC_SAMPLING'. The misestimate can concern several kind of operations: single table access, join, group by etc.

The state is 'NEW': the directive has been created but without more information about statistics.

The note gives an idea about the object concerned, which is detailed in the following view.

```
SQL> select object_name,subobject_name,object_type,notes from dba_sql_plan_dir_objects where owner='DEMO' ;

OBJECT_NAME          SUBOBJECT_NAME       OBJECT_TYPE          NOTES
-------------------- -------------------- -------------------- ---------
DEMO_TABLE           A                    COLUMN
DEMO_TABLE           B                    COLUMN
DEMO_TABLE           C                    COLUMN
DEMO_TABLE           D                    COLUMN
DEMO_TABLE                                TABLE
<obj_note>
 <equality_predicates_only>YES</equality_predicates_only>
 <simple_column_predicates_only>YES</simple_column_predicates_only>
 <index_access_by_join_predicates>NO</index_access_by_join_predicates>
 <filter_on_joining_object>NO</filter_on_joining_object>
</obj_note>
```

What is very interesting here is that the directive is not assigned to a statement, but only with a combination of tables, columns and kind of predicates.

Here the directive can be used by any statement on the DEMO_TABLE where columns A,B,C,D are involved.

Usable SQL Plan directive

What happens if I run the same query again?

```
SQL> select count(*) count1 from DEMO_TABLE where a=0 and b=0 and c=0 and d=0;

              COUNT1
--------------------
                 500

SQL> select * from table(dbms_xplan.display_cursor(null,null,'report iostats last'));

PLAN_TABLE_OUTPUT
--------------------------------------------------------------------------------
SQL_ID  86z3n2y70wavc, child number 1
-------------------------------------
select count(*) count1 from DEMO_TABLE where a=0 and b=0 and c=0 and d=0

Plan hash value: 1839825454
---------------------------------------------------------------------
|   Id | Operation            | Name       | Starts | E-Rows| A-Rows|
---------------------------------------------------------------------
|    0 | SELECT STATEMENT     |            |      1 |       |      1|
|    1 |  SORT AGGREGATE      |            |      1 |      1|      1|
|*   2 |   TABLE ACCESS FULL  | DEMO_TABLE |      1 |    500|    500|
---------------------------------------------------------------------

Predicate Information (identified by operation id):
---------------------------------------------------
   2 - filter(("A"=0 AND "B"=0 AND "C"=0 AND "D"=0))

Note
----
   - statistics feedback used for this statement
```

Thanks to automatic reoptimization the statement has been hard parsed again, now using the cardinality feedback (which is called statistics feedback in 12c) to get accurate estimations.

We have a new child number:

```
SQL> select sql_id,child_number,is_reoptimizable from v$sql where sql_id='86z3n2y70wavc';

SQL_ID                CHILD_NUMBER I
------------- -------------------- -
86z3n2y70wavc                    0 Y
86z3n2y70wavc                    1 N
```

And the internal state of the SQL Plan directive has changed from NEW to MISSING_STATS because the cardinality misestimate has been determined to be caused by missing stats (for correlated columns).

```
SQL> select directive_id,type,state,reason,created,last_modified,last_used,notes from dba_sql_plan_directives where
directive_id in(select directive_id from dba_sql_plan_dir_objects where owner='DEMO' );
        DIRECTIVE_ID TYPE            STATE      REASON
-------------------- ---------------- ---------- ------------------------------------
NOTES
--------------------------------------------------------------------------------------
 1350980939868665098 DYNAMIC_SAMPLING USABLE     SINGLE TABLE CARDINALITY MISESTIMATE
<spd_note>
  <internal_state>MISSING_STATS</internal_state>
  <redundant>NO</redundant>
  <spd_text>{EC(DEMO.DEMO_TABLE)[A, B, C, D]}</spd_text>
</spd_note>
```

Note that in 12.1.0.2 the state will show USABLE for both 'NEW' and 'MISSING STATS' and we have to check the NOTES column to get the internal state. It has changed from 2.1.0.1 where the state has all the details.

So, by executing the same statement, the SQL Plan Directive was not used (the statement was to be reoptimized anyway) but has been updated to validate its state (missing statistics).

Previously I said that SQL Plan Directive are also used with any statement that uses the same table. Let's change the statement (here changing count by max):

```
SQL> select max(a) from DEMO_TABLE where a=0 and b=0 and c=0 and d=0;

              MAX(A)
--------------------
                   0

SQL> select * from table(dbms_xplan.display_cursor(null,null,'report iostats last'));

PLAN_TABLE_OUTPUT
---------------------------------------------------------------------
SQL_ID  crqtabczzmmdv, child number 0
-------------------------------------
select max(a) from DEMO_TABLE where a=0 and b=0 and c=0 and d=0

Plan hash value: 1839825454

---------------------------------------------------------------------
|   Id  | Operation           | Name       | Starts | E-Rows | A-Rows |
---------------------------------------------------------------------
|    0  | SELECT STATEMENT    |            |      1 |        |      1 |
|    1  |  SORT AGGREGATE     |            |      1 |      1 |      1 |
|*   2  |   TABLE ACCESS FULL | DEMO_TABLE |      1 |    500 |    500 |
---------------------------------------------------------------------

Predicate Information (identified by operation id):
---------------------------------------------------
   2 - filter(("A"=0 AND "B"=0 AND "C"=0 AND "D"=0))

Note
-----
   - dynamic statistics used: dynamic sampling (level=2)
   - 1 Sql Plan Directive used for this statement
```

Then, for the same predicates – but different query- we have good estimations event for the first executions. And the dbms_xplan notes explains that:

- The SQL Plan Directive told the optimizer that some missing statistics can lead to misestimate with those predicates
- Then the optimizer has chosen to gather more stats with Dynamic Sampling

This is the first very nice feature of SQL Plan Directives: the optimizer knows that it will need to do more work in order to have good estimations.

DBMS_STATS

This is good, but not enough. Using Dynamic Sampling is only a short term solution. The durable way to solve the issue is to gather extended statistics for those correlated columns. And this is where SQL Plan Directives are even nicer: dbms_stats will check the directives and will gather the required statistics automatically.

Let's gather the status as usual, without creating any extended statistics ourselves:

```
SQL> exec dbms_stats.gather_table_stats(user,'DEMO_TABLE');

PL/SQL procedure successfully completed.
```

And see what has been gathered:

```
SQL> select extension_name,extension from user_stat_extensions where table_name='DEMO_TABLE';

EXTENSION_ EXTENSION
---------- --------------------------------
SYS_STSPJN ("A","B","C","D")
```

We have now statistics on those columns, because dbms_stats knew that they were missing when optimizing some queries.

For the moment, the directive state has not changed. But let's run another query (I've changed max to min):

```
SQL> exec dbms_spd.flush_sql_plan_directive;

PL/SQL procedure successfully completed.

SQL> select directive_id,type,state,reason,created,last_modified,last_used,notes from dba_sql_plan_directives where
directive_id in(select directive_id from dba_sql_plan_dir_objects where owner='DEMO' );

        DIRECTIVE_ID TYPE             STATE      REASON
-------------------- ---------------- ---------- ------------------------------------
NOTES
--------------------------------------------------------------------------------
 1350980939868665098 DYNAMIC_SAMPLING SUPERSEDED SINGLE TABLE CARDINALITY
<spd_note>
  <internal_state>HAS_STATS</internal_state>
  <redundant>NO</redundant>
  <spd_text>{EC(DEMO.DEMO_TABLE)[A, B, C, D]}</spd_text>
</spd_note>
```

The internal state is now 'HAS_STATS', the displayed state being 'SUPERSEEDED' which means that the problem has been fixed: we have enough statistics now.

Why did we need another execution to get that state? Because the optimizer has checked that the issue is now fixed permanently. If it were not the case, then the internal state would have been 'PERMANENT', meaning that the misestimate is permanent and the optimizer cannot rely on the gathered extended statistics.

Conclusion

That new component helps to share and store what the optimizer observes at execution time. It is intelligence and memory that are added to the cardinality feedback features. Those observation lead to dynamic sampling as a short term solution and to extended statistics gathering for long term solutions. When an issue is fixed (and the optimizer checks that as well) then the SQL Plan directives are automatically purges after one year. ■

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Miguel Anjo, Trivadis

7 significant changes to user access control and privileges in Oracle 12.1

Oracle 12c is bringing several new ways to ensure the security of the databases we manage. This short article describes seven changes related to users' privileges and access that appeared in releases 12.1.0.1 and 12.1.0.2.

1. READ and READ ANY TABLE privileges

Since long there was **SELECT, INSERT, UPDATE** and **DELETE** privileges over tables. With Oracle 12.1.0.2 there is the new **READ** and **READ ANY TABLE** privilege. The reason is that with the **SELECT** privilege the user is able to do **"LOCK TABLE xxx IN EXCLUSIVE MODE"** and **"SELECT … FOR UPDATE"** and therefore locking table for other users.

For this reason, once you migrate to Oracle 12c, it might be interesting to replace the **SELECT** privileges with **READ** privileges, like on read only users or roles.

Care should be taken for some strict situations as the READ privilege is not compliant with SQL92 security standard. Within this norm, an UPDATE or DELETE privilege requires in addition the SELECT privilege in order to perform DDL operations. Databases running Oracle Database Vault or having the SQL92_SECURITY parameter set to TRUE (the default is FALSE) should take particular attention.

2. Attach roles to PL/SQL program units – code based access control

The feature mentioned in this paragraph could give a full length article. Let's try to make it simple. As of version 12.1 it is possible to assign roles to PL/SQL code like functions, procedures and packages. This gives the possibility of raising user privileges only during the run time of the program, which is a big advantage for running code with invokers' rights (**authid current_user**). Until Oracle 11g for achieving the same there were two options: to grant the privileges directly to the user executing the procedure; or to run with definer rights – which in turn could be much more powerful than the necessary – and it would lose the caller environment information.

Let's see an example of how useful is this new feature. Imagine user **BERN** writes the following procedure to be used by other users:

```
CREATE OR REPLACE PROCEDURE bern.proc_current_user
  AUTHID CURRENT_USER
  AS
  BEGIN
    INSERT INTO bern.table1 (user, date)
      VALUES (SYS_CONTEXT('USERENV', 'CURRENT_USER'));
    COMMIT;
  END;
/

GRANT EXECUTE ON bern.proc_current_user TO geneva;

CREATE ROLE r_insert;
GRANT INSERT ON bern.table1 TO r_insert;
```

Until Oracle 11g the user **BERN** needs to give the following permissions:

```
GRANT r_insert TO geneva;
```

This means that as soon as user **GENEVA** logins, he has full insert privileges on **bern.table1** which is not secure in most situations.

From Oracle 12c it is now possible to user **BERN** to pass the privileges only to the procedure:

```
GRANT r_insert TO PROCEDURE bern.proc_current_user;
```

In order to help visualize the security context during a session within the different options of the example one can look at the following diagram.

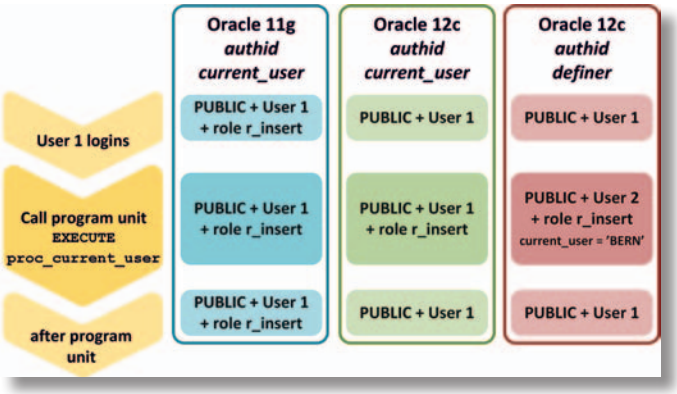

Figure 1 – Security context changes during call to program unit

It is visible that by using this new feature it is possible to keep to a minimum the privileges assigned to a user in a permanent fashion while not losing on functionality.

Oracle's idea behind this new possibility is to segregate the roles between the person who develops the code, the one who deploys the application and a privilege manager. It is the privilege manager the responsible for creating the necessary roles and granting them to the piece of code.

3. DELEGATE OPTION for roles

In order to assign existing roles to program units, a user needs to have GRANT ANY ROLE privilege or the ADMIN option for role. Since 12.1.0.2 a new grant option exists – DELEGATE – specifically for code based access control. As an example, privilege manager user ZURICH creates a role and grants to BERN which should then be able to grant it to a program unit:

```
CREATE ROLE r_insert;
GRANT SELECT, UPDATE ON bern.table1 TO r_insert;
GRANT r_insert TO bern WITH DELEGATE OPTION;
```

Then BERN has now the right to delegate the role to a program unit:

```
GRANT r_insert TO PROCEDURE bern.proc_current_user;
```

4. Proxy only connect user property

This yet undocumented feature allows to define application schemas which can only be accessed through a proxy user. It makes a very useful to assure that no user connects directly to the application schema, even by knowing its password.

Here how it works:

```
SQL> CREATE USER app_user IDENTIFIED BY xyz;
User created.

SQL> GRANT CREATE SESSION TO app_user;
Grant succeeded.

SQL> ALTER USER app_user PROXY ONLY CONNECT;
User altered.

SQL> CREATE USER personal_user IDENTIFIED BY prx1;
User created.

SQL> ALTER USER app_user GRANT CONNECT THROUGH personal_user;
User altered.

SQL> CONNECT app_user/xyz;
ERROR:
ORA-28058: login is allowed only through a proxy

SQL> CONNECT personal_user[app_user]/prx1;
Connected.

SQL> SELECT user FROM dual;
USER
------------------------------
APP_USER
```

As usual, the use of undocumented features are not supported by Oracle. The syntax to rollback the change is:

```
SQL> ALTER USER app_user CANCEL PROXY ONLY CONNECT;
```

5. RESOURCE role without UNLIMITED TABLESPACE privilege

While it is not recommended by Oracle to continue to use the RESOURCE role ("this role might not be created automatically by future releases of Oracle Database" as is written in the documentation), there is an important change on Oracle 12.1 – there is no more the UNLIMITED TABLESPACE system privilege.

The consequence of this change is that, when you grant the RESOURCE role, you need also to explicitly grant some quota on a tablespace to the user.

Imagine the common (not recommended) way to create users and a table:

```
SQL> CREATE USER bern IDENTIFIED BY xyz;
SQL> GRANT CONNECT, RESOURCE TO bern;

SQL> CONNECT bern/xyz
SQL> CREATE TABLE t1(c1 NUMBER);
```

Up to 11.2.0.4 you could do after as user 'BERN':

```
SQL> INSERT INTO t1 VALUES (1);
1 row created.
```

However from 12.1 this is what happens:

```
SQL> INSERT INTO t1 VALUES (1);
INSERT INTO t1 VALUES (1)

ERROR at line 1:
ORA-01950: no privileges on tablespace 'USERS'
```

6. SELECT ANY DICTIONARY protects critical SYS schema tables

As DBA we know that hashed passwords can be found at **USER$** table. Some other tables like **DEFAULT_PWD$, ENC$, LINK$, USER_HISTORY$, CDB_LOCAL_ADMINAUTH$** and **XS$VERIFIERS** also contain very sensitive information.

Starting with Oracle 12c, the **SELECT ANY DICTIONARY** role does not include access to those views, limiting only to SYS user the access to them.

7. Last login information

Another small improvement that can greatly help to keep an environment safe is the information about the last successful login. It permits, without any other auditing, to know when an account was last used. This allows to make the DBA sure when he can delete an unused account and also can make the user aware if someone else abused of his account.

This information is visible at the new **LAST_LOGIN** column of the **DBA_USERS** view.

Logging in to SQL*Plus the user can also see when the last successful login happen:

```
$sqlplus bern/xyz

SQL*Plus: Release 12.1.0.1.0 Production on Thu Oct 30 12:38:25 2014
Copyright (c) 1982, 2013, Oracle.  All rights reserved.
Last Successful login time: Thu Oct 30 2014 12:28:32 +01:00
```

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Testing Delphix: easy data virtualization

File system technologies such as snapshots, cloning, copy-on-write are arising to handle the ever increasing size of databases and the multiplying number of required copies (backups, preprod, test environment). But there are new technologies to learn beyond the basic storage system options. EMC, NetAPP, Oracle ZFS Storage Appliance are hardware appliance and requires specialized skills. But there is a new breed of products that have arisen to tackle both the technical obstacle as well as the specialized skills obstacle. The new product is a software appliance: it runs as a VMware virtual machine and exposes the virtual storage through NFS.
The software is intimately integrated with the database providing power and ease of use. At Oracle Open World I tested it and discussed with the oracle experts Kyle Hailey (Ace and Oaktable) and Tim Gorman (Ace Director and Oaktable) that are now working at Delphix.

Why data virtualization?

What is the size of your database? Let's say you have a small to medium size – 200GB. And you generate 20GB of archivelog every day. Let's go into production needs first. You requires to keep 30 days of backup. With daily full backups that's more than 5TB of backupsets. If you do weekly full backups and daily incrementals, you may reduce it to 2TB.

Ok 2TB for backups. Now you need environments for preproduction (testing upgrades), UAT (testing releases), qualification (testing performance). Your UAT users dream about having their own database for each testers, so they don't overlap on data.

And now about developers. If you want them to release efficient application you should give them an environment with representative volume of data. And once again each developer team has its own set of data set. How many database do you need to provide?

That requirements exists for years. But today the delivery delays have decreased a lot. We must be agile. Developers are committing to continuous integration. They have a bunch of test cases to run on a fixed set of data. And they need to rewind the database each time.

I forget one environment: bugfix. What if you find an application bug on production? You have a patch but want to test it. If you have Active Data Guard (i.e. Enterprise Edition plus ADG option), you can use one of the physical standby to do a snapshot standby. What if you were able to just click on a GUI and provision a new copy instantaneously?

Well, it's a bit more than a click. It is a click and slide:

Yes: you can choose the point in time snapshot that you want to copy.

Source databases

So you have your existing databases and they can act as source for cloning. You don't have to do anything about them except installing the Delphix Toolkit on their server.

Here is the screen to define the source server. Toolkit deployment is automatic:

You just set the host connection and where to install the Toolkit.

Then you add the databases.

How does it works? Delphix will use RMAN to get the source database.

The first time you add the source database, it will do a full backup through RMAN. And only the first time. Then only incremental changes are collected (using incremental backups and/or archived logs) to maintain the copy on the Delphix server.

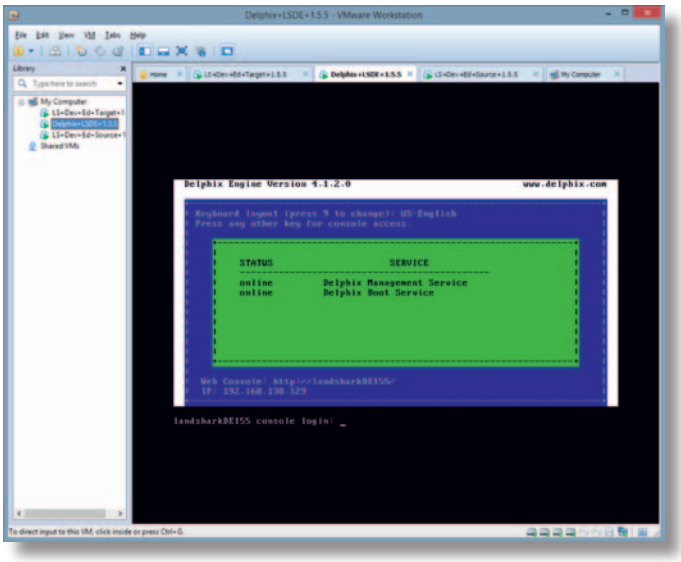
The Delphix server is a VMware virtual machine. You allocate storage as you want and Delphix maps its own file system onto the storage. Delphix uses a copy-on-write filesystem in order to be able to rewind to any state between the first initialization (or the defined window) and the current time. In addition to that the storage is compressed by Delphix. Then the copy is 2 or 3 times smaller than the source database and it is able to flashback in the past.

Clone databases

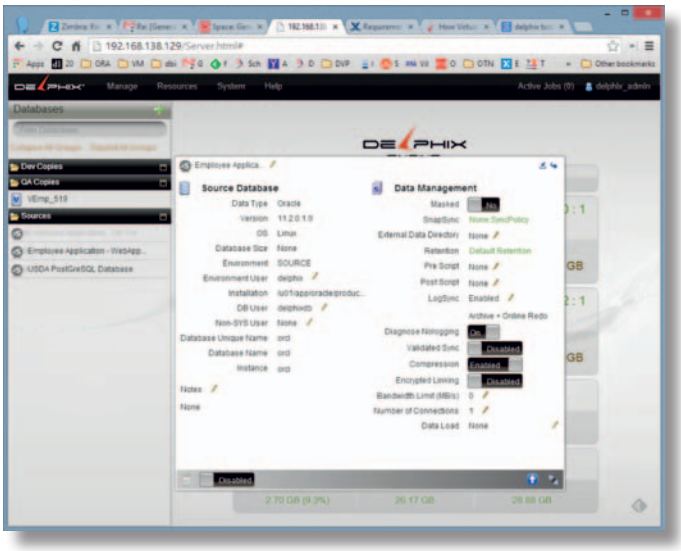
Then you have the base: a copy of your production which you can view as of any timestamp.

And with a few clicks you can provision a clone immediately: data blocks are not copied until they are changed.

I have the '#cloneattack' environment on my laptop. Delphix is running and I can connect with any web browser to get the full interface, The server running in VMware Workstation:

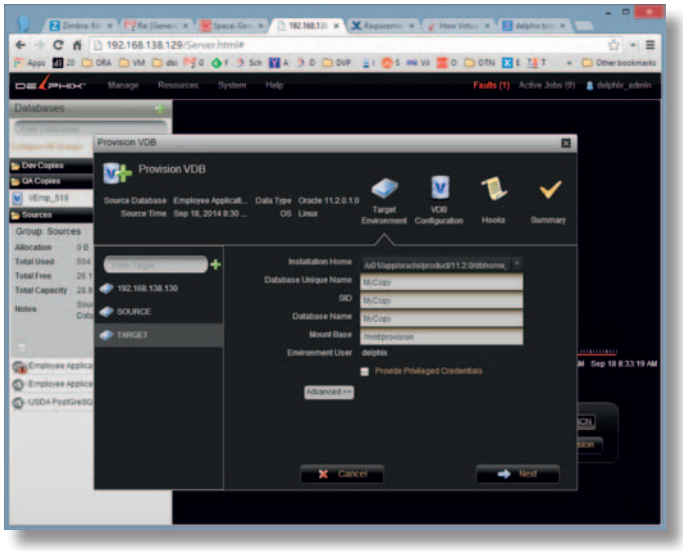


And the console, connected from my browser. Here I show the source database properties. Delphix has a nice GUI: you click on the upper right arrow to flip between summary and detail:

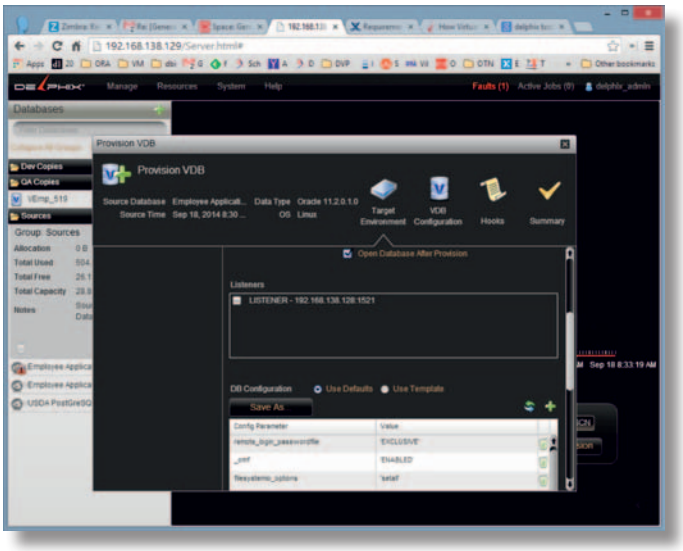


Ok, now I want to get a copy of it – as of 08:30 this morning:

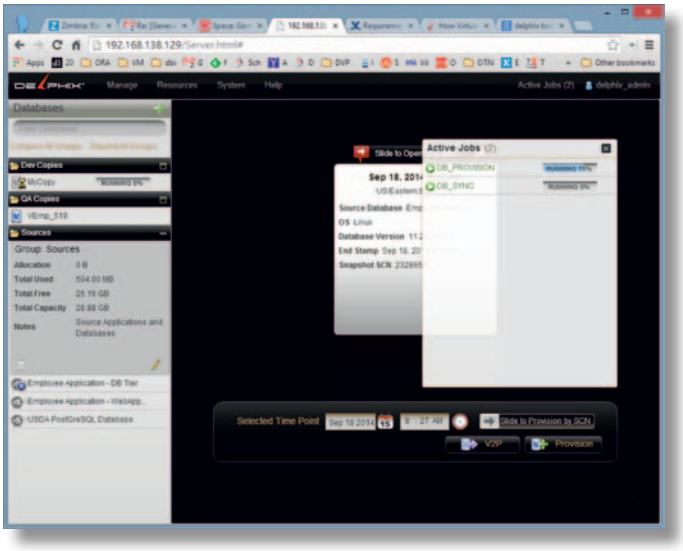
I have then to define where to put it (storage is virtualized but I need a server to run the instance) and a name for it:



It will create automatically an instance for it (I need the Delphix toolkit installed on the target server as well) and I can change the default init.ora parameters, file location, listener etc.



Ok, after a few screens Delphix is doing everything automatically. I can control the running jobs: DB_PROVISION creates my instance and database and DB_SYNC manages the data virtualization to show the data from the source point-in-time I have chosen.



How long does it take? That's just a few minutes and the database is opened in read write, ready to accept connections with the connection string defined by the listener and database configuration.

You can see it on the target server:

```
[oracle@linuxtarget ~]$ ps -edf | grep pmon
oracle   57363     1  0 02:19 ?        00:00:00 ora_pmon_VEmp519
oracle   60795     1  0 03:29 ?        00:00:00 ora_pmon_MyCopy
oracle   61180 61147  0 03:33 pts/0    00:00:00 grep pmon
```

and the listener is listening on it:

```
[oracle@linuxtarget ~]$ lsnrctl status
LSNRCTL for Linux: Version 11.2.0.1.0 - Production on 30-SEP-2014 03:34:01
Copyright (c) 1991, 2009, Oracle.  All rights reserved.
Connecting to (DESCRIPTION=(ADDRESS=(PROTOCOL=TCP)(HOST=linuxtarget.delphix.local)(PORT=1521)))
STATUS of the LISTENER
------------------------
Alias                     LISTENER
Version                   TNSLSNR for Linux: Version 11.2.0.1.0 - Production
Start Date                29-SEP-2014 21:49:14
Uptime                    0 days 5 hr. 44 min. 47 sec
Trace Level               off
Security                  ON: Local OS Authentication
SNMP                      OFF
Listener Parameter File   /u01/app/oracle/product/11.2.0/dbhome_1/network/admin/listener.ora
Listener Log File         /u01/app/oracle/diag/tnslsnr/linuxtarget/listener/alert/log.xml
Listening Endpoints Summary...
  (DESCRIPTION=(ADDRESS=(PROTOCOL=tcp)(HOST=linuxtarget)(PORT=1521)))
Services Summary...
Service "MyCopy" has 1 instance(s).
  Instance "MyCopy", status READY, has 1 handler(s) for this service...
Service "VEmp_519" has 1 instance(s).
  Instance "VEmp519", status READY, has 1 handler(s) for this service...
The command completed successfully
```

I like to see the structure of the database with RMAN 'report schema':

```
RMAN> report schema;

using target database control file instead of recovery catalog
Report of database schema for database with db_unique_name MYCOPY

List of Permanent Datafiles
===========================
File Size(MB) Tablespace  RB segs  Datafile Name
---- -------- ---------  -------  ------------------------
1    680      SYSTEM         ***  /mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/system01.dbf
2    530      SYSAUX         ***  /mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/sysaux01.dbf
3    60       UNDOTBS1       ***  /mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/undotbs01.dbf
4    38       USERS          ***  /mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/users01.dbf

List of Temporary Files
=======================
File Size(MB) Tablespace           Maxsize(MB)  Tempfile Name
---- -------- -------------------- -----------  --------------------
1    20       TEMP                 32767        /mnt/provision/MyCopy/temp/temp1.dbf
```

We see that the datafiles are on the NFS mountpoint, with the name of the copy and the name of the source as well.

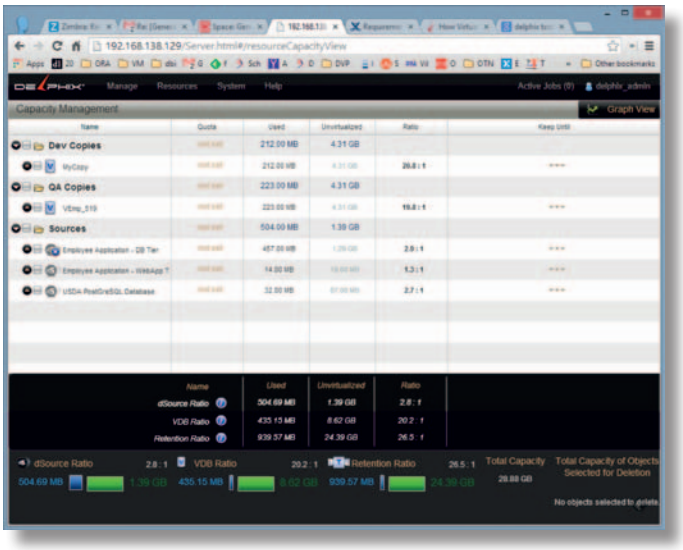
And finally I like to see all what happened on the background:

```
[oracle@linuxtarget trace]$ cat alert_MyCopy.log
Tue Sep 30 03:26:01 2014
Starting ORACLE instance (restrict)
LICENSE_MAX_SESSION = 0
LICENSE_SESSIONS_WARNING = 0

…
System parameters with non-default values:
  processes                    =  150
  event                        =  "10052 trace name context forever"
  filesystemio_options         =  "setall"
  _omf                         =  "DISABLED"
  control_files                =  "/mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/control01.ctl"
  _controlfile_enqueue_timeout =  4500
  db_block_size                =  8192
  compatible                   =  "11.2.0.0.0"
  log_archive_dest_1           =  "location=/mnt/provision/MyCopy/source-archive MANDATORY"
  log_archive_format           =  "arch_%t_%s_%r.log"
  db_create_file_dest          =  "/mnt/provision/MyCopy/datafile"
  remote_login_passwordfile    =  "EXCLUSIVE"
  audit_sys_operations         =  FALSE
  job_queue_processes          =  0
  _system_trig_enabled         =  FALSE
  audit_trail                  =  "NONE"
  db_name                      =  "orcl"
  db_unique_name               =  "MyCopy"
  open_cursors                 =  300
  _diag_hm_rc_enabled          =  FALSE
…
ALTER DATABASE   MOUNT
…
alter database rename file
'/u02/app/orcl/data/u01/app/oracle/oradata/orcl/system01.dbf'
,'/u02/app/orcl/data/u01/app/oracle/oradata/orcl/sysaux01.dbf'
,'/u02/app/orcl/data/u01/app/oracle/oradata/orcl/undotbs01.dbf'
,'/u02/app/orcl/data/u01/app/oracle/oradata/orcl/users01.dbf'
 to
'/mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/system01.dbf'
,'/mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/sysaux01.dbf'
,'/mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/undotbs01.dbf'
,'/mnt/provision/MyCopy/datafile/u02/app/orcl/data/u01/app/oracle/oradata/orcl/users01.dbf'
```

What I see here is that this has been designed by people that really knows how oracle works in depth. It's not just a simple clone script that works only in the common case. It's real DBA stuff.

Then what do you have?



You have a database that takes minimal storage and is seen as an exact point-in-time copy of the source database.

You can use it, rewind it to another point in time, you can refresh it from source, you can clone it to another copy etc.

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Use Cases

So what can you do with virtual databases?

- You can give a database to each tester, each developer and they have a full volume database without taking too much storage
- You can give a database for continuous integration that can be rewind before each test (Delphix has an API to control without the GUI)
- You set up QA environments with real data. Choosing a point-in-time let open a database in an 'end-of-month' situation for example if you want to test a monthly job.
- You can have synced databases for protection. If the Delphix VM is set with high availability, then you can get an exact copy of a production database that is not available anymore.
- You can quickly provision a database for bugfix, showing an exact situation in the past where you had the bug happening.
- You can use it for backups: you reduce storage and reduce RPO and RTO. Do you prefer RMAN backups as it detects block corruption? Why not automating a daily clone provisioning that you open and run a rman validate database on it?
- You can avoid lot of the storage you use for backup retention. You can get yearly snapshots to keep 10 years, monthly snapshots to keep 2 years and be able to get a copy from any point-in time from the previous two months.
- Your BI ODS database can be refreshed with Delphix in a few minutes. Remember: Delphix maintains its copy of the production database (synchronizing with incrementals and archivelogs) so you don't impact production performance even with high activity on the clone.

Those are the few Use Cases that are the most common I see in many companies.

Conclusion

I've heard of Delphix for months, mainly because people at Delphix are well known in the Oracle community. And Jonathan Lewis has presented that solution on several occasions. His performance tuning activity often show the need to have an exact copy of production at a specific state in order to reproduce the issue.

But I like to learn about a product by testing it instead of reading docs or looking at presentations. And the #cloneattack organized at the Oracle Open World was a good occasion for that. By playing with it just a few hours, I can see how it addresses some common questions frequently raised by my customers. Both in operation and development teams.

This article is only about what I've tested. There is a lot more. Delphix is not only for Oracle. It does the same for Microsoft SQL Server. PostgreSQL is also supported. ■

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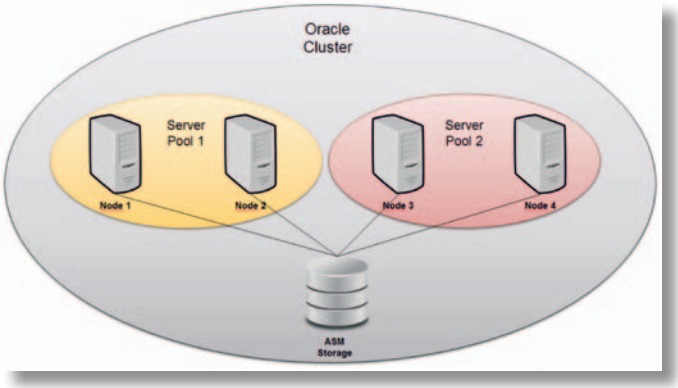
Implementing policy-managed Oracle databases

Policy-managed databases appeared with Oracle 11g R2. However, they are not very common yet, even if Oracle now presents policy-databases as the default installation mode and as a best practice. Anyway, this is quite interesting to know more about this feature which fully automatizes the databases repartition within an Oracle cluster. In this article, I will describe how to implement and handle policy-managed databases. I will also present a new Oracle 12c feature which allows to adapt the cluster resources to the workload requirements on the fly.

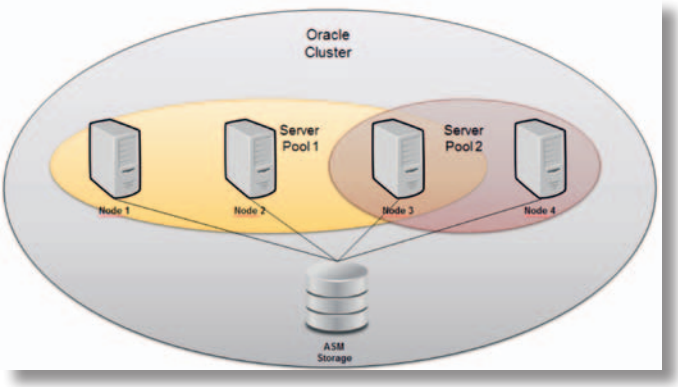
Understanding the server pool concept

The main concept to know and to understand when we speak about policy-managed databases is the one of server-pool. A server pool is a logical group of servers within the cluster, sharing resources for databases and/or services. The following schema shows two server pools, having two nodes each.



Schema 1

The schema 1 shows a simple configuration, where a server is member of one and only one server pool. However, a node can be part of one or multiple server pools, depending on the configuration. The following configuration (Schema 2) is also a valid one:



SMS >>>

Database Ideas

Mit dem neuen Forum «Database Ideas» innerhalb des Oracle Technology Network (OTN) bietet Oracle allen Anwendern von nun an die Möglichkeit, sich aktiv an der Weiterentwicklung der Oracle-Datenbank zu beteiligen. Verbesserungsvorschläge, Änderungswünsche und Anregungen können dort zur Diskussion gestellt werden. Damit reagiert Oracle endlich auf den lang gehegten Wunsch der Datenbank-Community der DOAG unter der Leitung von Christian Trieb.

Mehr unter: http://tinyurl.com/mjg2kad

Here we can see that Node 3 is member of both Server pool 1 and server pool 2. It is a candidate to move in either server pool 1 or server pool 2, while node 1 and 2 are physically linked to the first server pool and node 4 to the second server pool.

We can control that by using a “Candidate Servers” list when creating the server pool.

Concerning the database or the service running in a policy managed context, with server pools, the database instance is not assigned anymore to a named server, but to a server pool. The cluster will manage the instance placement over the nodes according to the performances requirements.

Scalability and availability of applications (databases, services...) within a server-pool can be dynamically defined using MIN_SIZE, MAX_SIZE and MAXIMUM parameters. These parameters are defined at server pool level:

- **MIN_SIZE** represents the minimal number of nodes which should compose the server-pool. If the number of nodes falls below the MIN_SIZE value, Oracle moves servers from other server-pools – if available – to satisfy this parameter.
- **MAX_SIZE** represents the maximal number of nodes available in a server-pool. MAX_SIZE can be considered as a kind of target or as a “must have". Oracle will satisfy this parameter only, if all server-pool MIN_SIZE are reached first.
- **IMPORTANCE** controls the availability of an application. Its value can range from 0 (not important) to 1000. The server pool with the higher importance value will have resources allocated in priority: new available nodes are assigned to the most important server-pool first and nodes can be de-assigned from a less important server-pool to satisfy the MIN_SIZE of the most important server-pool.

As you may expect, the IMPORTANCE parameter will be used to guarantee availability of business critical services, to the detriment of lower important applications, such as development.

Creating a server-pool

In this example, I will use a cluster composed of four identic nodes. Two are up and running, the two remaining have cluster services not running at this time.

First, I will create two server-pools. One for production databases, named **srvprod** and one for development/test databases, named **srvtest**. As a first example, I set MIN and MAX values at 1 and IMPORTANCE value at 0, the minimum. I use the srvctl tool as oracle user.

As described earlier on this post, we can create a server pool by specifying a server list. If we specify one or several nodes as part of a server pool, only the listed nodes would be able to migrate to this server pool. We speak about “Candidate server names". The candidate list implicitly excludes all non-mentioned server names, unless you leave this list empty.

Indeed, if you do not specify any node name in the candidate list, all nodes are considered to be able to migrate to the server pool. It means that all nodes are “candidates” for this pool.

Here I will not specify candidate servers, so all nodes within my cluster can migrate to any server pools. Note that it would have the same effect as specifying the four nodes as candidate for each server pool.

```
[oracle@node01 ~]$ srvctl add srvpool -g srvtest -l 1 -u 1 -i 0
[oracle@node01 ~]$ srvctl add srvpool -g srvprod -l 1 -u 1 -i 0
```

This is the list of options used:

- -g: name of the server pool
- -l: minimum number of nodes
- -u: maximum number of nodes
- -i: importance value

We can get the status of the different server pools with the following command:

```
[oracle@node01 ~]$ srvctl status srvpool -detail
Server pool name: Free
Active servers count: 0
Active server names:
Server pool name: Generic
Active servers count: 0
Active server names:
Server pool name: srvprod
Active servers count: 1
Active server names: node02
NAME=node02 STATE=ONLINE
Server pool name: srvtest
Active servers count: 1
Active server names: node01
NAME=node01 STATE=ONLINE
```

I have created only two server pools. However, we find two additional server pools in the output above: a “FREE” server pool and a “GENERIC” server pool. In fact, these pools are default pools created by Oracle which always exist no matter policy managed is enabled or not.

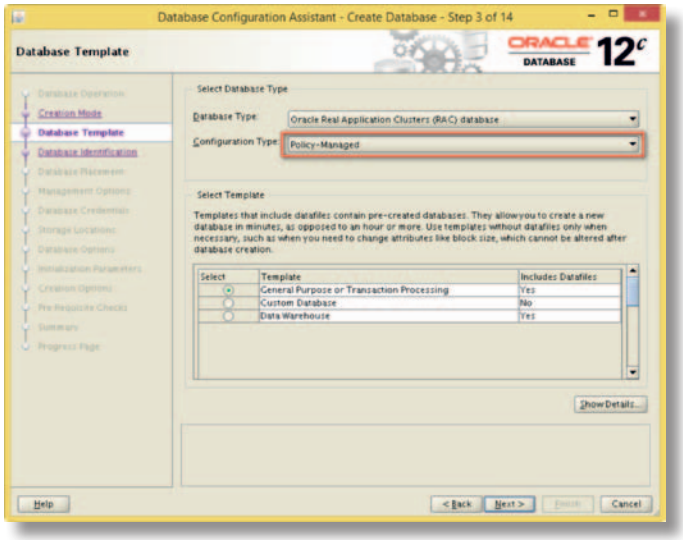
- The free server-pool contains unassigned servers. Servers migrate here first before being assigned to a server pool. The FREE pool is empty since the two available servers of my example are already assigned in srvprod and srvtest pools.
- The generic server pool contains servers running admin-managed databases. When policy-managed database is not used, all nodes of the cluster are member of the GENERIC pool. Here it is empty because there is no admin-managed database running in this cluster.

Assigning a database to a server-pool

We can assign a new database to a server-pool with dbca or with command line.

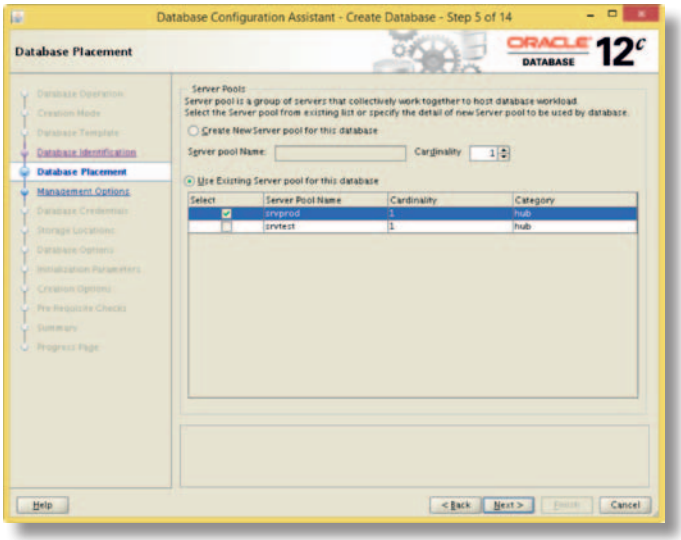
DBCA

In the following example, I have created the DB1 database in the existing srvprod server-pool. Policy managed is the default choice proposed starting with Oracle 12c:



DBCA gives the choice to create a new server pool to assign the new database, or use an existing server pool. Accessible server pools are listed in the following window. The cardinality represents the number of nodes available on the server pool.

By creating a new server pool from DBCA with a given cardinality, DBCA will create the server pool using N nodes available from the Free server pool, where N is the user provided cardinality.



Once the database is created, we can see that it has started on available nodes of the srvprod server-pool:

```
[oracle @ node01 ~]$ srvctl status database -d DB1
Instance DB1_1 is running on node node01
```

As for an admin managed database, the instance gets a suffix number but the name is automatically generated by Oracle and an “underscore” character is used with policy managed to define the name of the instance. Here DB1 gets DB1_1 instance name.

I have created a second database DB2 in the srvtest server-pool.

```
[oracle@node01 ~]$ srvctl status database -d DB2
Instance DB2_1 is running on node node02
```

Impact of server-pool parameters

Let's study the impact of MIN_SIZE, MAX_SIZE and IMPORTANCE parameters. In this example, I will first change the value of these parameters for the srvprod server pool and then I will start cluster services on the two nodes still powered off.

The srvprod pool will have a MIN_SIZE of 2 and a MAX_SIZE of 3, so Oracle will guarantee srvprod to have at least two nodes running. Pool srvtest remains with 1 for both MIN and MAX size.

```
[oracle@node01 ~]$ srvctl modify srvpool -min 2 -max 3 -serverpool srvprod
```

We have now the following configuration:

```
[oracle@node01 ~]$ srvctl config srvpool
...
Server pool name: srvprod
Importance: 0, Min: 2, Max: 3
Category: hub
Candidate server names:
Server pool name: srvtest
Importance: 0, Min: 1, Max: 1
Category: hub
Candidate server names:
```

(Note that "Candidate server names" is empty since I have not defined any server as part of a server pool.)

Now I start cluster services on node 03 and node 04.

```
[root@node03 bin]# ./crsctl start crs
CRS-4123: Oracle High Availability Services has been started.
```

```
[root@node04 bin]# ./crsctl start crs
CRS-4123: Oracle High Availability Services has been started.
```

After few minutes, I see that srvprod has now 3 nodes, satisfying the MAX_SIZE parameter. Srvtest has still 1 node and it is conform to the MIN_SIZE and MAX_SIZE defined:

```
[oracle@node01 ~]$ srvctl status srvpool
...
Server pool name: srvprod
Active servers count: 3
Active server names: node01,node03,node04
NAME=node01 STATE=ONLINE
NAME=node03 STATE=ONLINE
NAME=node04 STATE=ONLINE
Server pool name: srvtest
Active servers count: 1
Active server names: node02
NAME=node02 STATE=ONLINE
```

The test database DB2 is still running on node 02:

```
[oracle@node01 ~]$ srvctl status database -d DB2
Instance DB2_1 is running on node node02
```

Example 1: MIN_SIZE and MAX_SIZE

Let's see what happens if I stop cluster services on node 02 which is the single server running in the srvtest pool, with DB2 instance running.

```
[root@node02 bin]# date
Mon Apr 28 11:49:09 CEST 2014[root@node02 bin]# ./crsctl stop crs
CRS-2791: Starting shutdown of Oracle High Availability Services-managed
resources on 'node02'
CRS-2673: Attempting to stop 'ora.crsd' on 'node02'
CRS-2790: Starting shutdown of Cluster Ready Services-managed resources
on 'node02'
CRS-2673: Attempting to stop 'ora.db2.db' on 'node02'
...
```

To see what is going on, I consult the crsd.log file on the master node of the cluster.

Tip

To identify the master node within a cluster, look at the "OCR MASTER" string value with a grep command on the crsd.log from any node. You will get "I AM THE NEW OCR MASTER" on the primary node, or "NEW OCR MASTER IS N" on secondary nodes, where N is the number of the node returned by 'olsnodes -n' command:

```
[grid@node01 ~]$ cat /u00/app/12.1.0/grid/log/node01/crsd/crsd.log | grep -i
'ocr master'
...
...
2014-04-28 04:53:16.388: [  OCRMAS][2831439616]th_master:12: I AM THE NEW
OCR MASTER at incar
1. Node Number 1

[grid@node01 ~]$ olsnodes -n
node01  1
node02  2
node03  3
node04  4
```

The crsd.log file in /u00/app/12.1.0/grid/log/node01/crsd/crsd.log shows following events:

```
2014-04-28 11:50:50.878: [    CRSPE][2296276736] {1:4797:769} Pool ora.srvtest [min:1][max:1][importance:0] NO SERVERS ASSIGNED
  cannot be a donor  for pool ora.srvtest [min:1][max:1][importance:0] NO SERVERS ASSIGNED
2014-04-28 11:50:50.878: [    CRSPE][2296276736] {1:4797:769}Server [node04] has been un-assigned from the server pool: ora.
  srvprod
2014-04-28 11:50:50.880: [    CRSPE][2296276736] {1:4797:769}Server [node04] has been assigned to the server pool: ora.srvtest
...
2014-04-28 11:50:50.881: [    CRSPE][2296276736] {1:4797:769} Server [node04] has changed state from [ONLINE] to [RECONFIGURING]
...
2014-04-28 11:50:50.882: [    CRSPE][2296276736] {1:4797:769} CRS-2673: Attempting to stop 'ora.db1.db' on 'node04'
...
2014-04-28 11:51:15.708: [    CRSPE][2296276736] {1:4797:769} CRS-2677: Stop of 'ora.db1.db' on 'node04' succeeded
2014-04-28 11:51:15.708: [    CRSPE][2296276736] {1:4797:769} Updated state details for server node04 from [STOPPING RESOURCES]
  to : []
2014-04-28 11:51:15.708: [    CRSPE][2296276736] {1:4797:769} Server [node04] has changed state from [RECONFIGURING] to [ONLINE]
...
2014-04-28 11:51:15.718: [    CRSPE][2296276736] {1:4797:769} CRS-2672: Attempting to start 'ora.db2.db' on 'node04'
...
2014-04-28 11:51:54.749: [    CRSPE][2296276736] {1:4797:769} RI [ora.db2.db 1 1] new internal state: [STABLE] old value: [STAR-
  TING]
2014-04-28 11:51:54.749: [    CRSPE][2296276736] {1:4797:769} RI [ora.db2.db 1 1] new external state [ONLINE] old value: [OFF-
  LINE] on node04 label = [Open]
2014-04-28 11:51:54.749: [    CRSPE][2296276736] {1:4797:769} Set LAST_SERVER to node04 for [ora.db2.db 1 1]
2014-04-28 11:51:54.750: [    CRSPE][2296276736] {1:4797:769} Set State Details to [Open] from [ Instance Shutdown] for [ora.db2.
  db 1 1]
2014-04-28 11:51:54.750: [    CRSPE][2296276736] {1:4797:769} CRS-2676: Start of 'ora.db2.db' on 'node04' succeeded
```

At the top of the output, we can see that Node 04 has been deassigned from srvprod server pool and has been assigned to srvtest serverpool. We can see that the prod instance of DB1, running on node04, has been shut down and then the dev DB2 instance has started.

According to the log file, Oracle has satisfied the MIN value of 2 for srvprod and the MIN value of 1 for srvtest.

We can confirm with srvctl commands:

```
[oracle@node01 crsd]$ srvctl status database -d DB2
Instance DB2_1 is running on node node04

[oracle@node01 ~]$ srvctl status srvpool
...
Server pool name: srvprod
Active servers count: 2
Active server names: node01,node03
NAME=node01 STATE=ONLINE
NAME=node03 STATE=ONLINE
Server pool name: srvtest
Active servers count: 1
Active server names: node04
NAME=node04 STATE=ONLINE
```

MIN_SIZE value remains satisfied for both srvprod and srvtest pools and both databases DB1 and DB2 are accessible.

Example 2: IMPORTANCE parameter

To demonstrate how this parameter works, I will edit properties for the srvprod server pool and set an IMPORTANCE of 1. Srvprod will become more important than srvtest, in case of failure.

I will then stop cluster services on node 03, which is part of srvprod server pool.

First, I set an IMPORTANCE value of 1 for the srvprod server pool:

```
[oracle@node01 ~]$ srvctl modify srvpool -importance 1 -serverpool srvprod
[oracle@node01 ~]$ srvctl config srvpool -serverpool srvprod
Server pool name: srvprod
Importance: 1, Min: 2, Max: 3
Category: hub
Candidate server names:
```

Now, I stop cluster services on node 03. The number of remaining nodes in the srvprod server pool will fall under the MIN_SIZE value, with only one server remaining.

```
[root@node03 bin]# ./crsctl stop crs
CRS-2791: Starting shutdown of Oracle High Availability Services-managed resources on 'node03'
CRS-2673: Attempting to stop 'ora.crsd' on 'node03'
CRS-2790: Starting shutdown of Cluster Ready Services-managed resources on 'node03'
...
CRS-4133: Oracle High Availability Services has been stopped.
```

Let's see what is happening in the crsd.log file:

```
2014-04-28 06:27:29.222: [    CRSPE][2296276736] {1:4797:910}Server [node04] has been un-assigned from the server pool: ora.
  srvtest
2014-04-28 06:27:29.223: [    CRSPE][2296276736] {1:4797:910}Server [node04] has been assigned to the server pool: ora.srvprod
...
2014-04-28 06:27:29.224: [    CRSPE][2296276736] {1:4797:910} Created alert : (:CRSPE00165:) :  Server Pool ora.srvtest has fal-
  len below minimum size
...
2014-04-28 06:27:29.224: [    CRSPE][2296276736] {1:4797:910} Server [node04] has changed state from [ONLINE] to [RECONFIGURING]
2014-04-28 06:27:29.224: [    CRSPE][2296276736] {1:4797:910} Evaluating stop of [ora.db2.db] on [node04] with parameters:
  Options:0x1;Identity: root;Reason:system;
...
2014-04-28 06:27:29.226: [    CRSPE][2296276736] {1:4797:910} CRS-2673: Attempting to stop 'ora.db2.db' on 'node04'
...
2014-04-28 06:28:00.352: [    CRSPE][2296276736] {1:4797:910} CRS-2677: Stop of 'ora.db2.db' on 'node04' succeeded
2014-04-28 06:28:00.352: [    CRSPE][2296276736] {1:4797:910} Updated state details for server node04 from [STOPPING RESOURCES]
  to : []
2014-04-28 06:28:00.352: [    CRSPE][2296276736] {1:4797:910} Server [node04] has changed state from [RECONFIGURING] to [ONLINE]
...
2014-04-28 06:28:00.360: [    CRSPE][2296276736] {1:4797:910} CRS-2672: Attempting to start 'ora.db1.db' on 'node04'
...
2014-04-28 06:28:25.563: [    CRSPE][2296276736] {1:4797:910} CRS-2676: Start of 'ora.db1.db' on 'node04' succeeded
```

The node 04, previously assigned to the pool srvtest to satisfy the MIN_VALUE of 1, is now re assigned to the srvprod pool. The corresponding dev DB2 database is powered off to allow the production DB1 to start. This behavior is caused by the IMPORTANCE parameter, greater for srvprod than for srvtest.

Srvtest pool has now no server running, meaning no instance or service available. But the production availability has been guaranteed:

```
[oracle@node01 ~]$ srvctl status srvpool
...
Server pool name: srvprod
Active servers count: 2
Active server names: node01,node04
NAME=node01 STATE=ONLINE
NAME=node04 STATE=ONLINE
Server pool name: srvtest
Active servers count: 0
Active server names:

[oracle@node01 ~]$ srvctl status database -d DB2
Instance DB2_1 is running on node node01
Instance DB2_2 is running on node node04

[oracle@node01 ~]$ srvctl status database -d DB2
Database is not running.
```

This example well demonstrates how the IMPORTANCE parameter may influence the availability of a database or a service within a policy-managed cluster.

Policy managed conclusion

With two examples, I have shown how MIN_SIZE, MAX_SIZE and IMPORTANCE parameters can impact the application availability within a policy-managed database cluster. It shows how much it is important to define these parameters according to your business rules, because it really helps to keep available resources for a critical business.

In a next post, I will show how we can use these parameters to dynamically adapt the available resources to the workload of an application.

Starting from Oracle 12c: Policies and policy sets

To conclude this post with a new feature shipped with Grid Infrastructure 12c, I will quickly present the concept of cluster configuration policies and policy sets. It allows the administrator to dynamically reconfigure all server pools at a time, with a single command, according to the business workload.

To take a simple example, imagine you have two servers allocated for your production instances and two servers allocated for you development instances. This repartition is conform to your daily workload. But you have very intensive batches running every night on your production databases, which makes production servers to be over loaded while no developers are using the testing environments… What a great resources wasting! So why do not reassign the testing resources to the production pool until the batch processes are finished? Simply because it is too fastidious to reconfigure the whole cluster to re assign some servers in a pool just for hours…

Oracle found the solution and now provides policies sets.

A policy set is a “document” (Oracle definition) which contains one or more policies. Each policy contains the configuration of all server-pools in the cluster for a given time: MIN_SIZE and MAX_SIZE, IMPORTANCE, server list etc. You can configure several policies and then switch between them according to your needs.

It will make the administrator able to activate a policy or another, according to the workload or business needs, to dynamically and immediately change all the configuration of the cluster server-pools. For instance: use two servers for production and two servers for development during working days and allocate the four servers for production during nightly batch processes.

There is a demonstration.

I have reset my configuration which is now as follows:

```
[oracle@node01 ~]$ srvctl config srvpool
Server pool name: Free
Importance: 0, Min: 0, Max: -1
Category:
Candidate server names:
Server pool name: Generic
Importance: 0, Min: 0, Max: -1
Category:
Candidate server names:
Server pool name: srvprod
Importance: 0, Min: 2, Max: 2
Category: hub
Candidate server names:
Server pool name: srvtest
Importance: 0, Min: 2, Max: 2
Category: hub
Candidate server names:
```

Two nodes are currently running in srvprod pool and two nodes running in srvtest pool.

The instance DB1 (prod) is running on nodes 1 and 4:

```
[oracle@node01 ~]$ srvctl status database -d DB1
Instance DB1_1 is running on node node01
Instance DB1_2 is running on node node04
```

The instance DB2 (dev) is running on nodes 2 and 3:

```
[oracle@node01 ~]$ srvctl status database -d DB2
Instance DB2_1 is running on node node02
Instance DB2_2 is running on node node03
```

A simple way to benefit of policies and policy sets is to create a text file with the wanted configuration. Following steps explain how to proceed.

Step 1: Create a new policy set

The crsctl tool offers an option to generate a new policy set file, based on the current active configuration. We must connect as grid:

```
[grid@node01 ~]$ mkdir -p /u00/app/oracle/admin/policies
[grid@node01 ~]$ crsctl create policyset -file /u00/app/oracle/admin/policies/new_policy
```

The file has been generated:

```
[grid@node01 ~]$ ll /u00/app/oracle/admin/policies
total 4
-rw-r--r-- 1 grid oinstall 305 Apr 29 16:21 new_policy
```

The file contains the current configuration, which corresponds to the DEFAULT policy:

```
[grid@node01 ~]$ cat /u00/app/oracle/admin/policies/new_policy
SERVER_POOL_NAMES=Free ora.srvprod ora.srvtest
POLICY
  NAME=Default
  SERVERPOOL
    NAME=ora.srvprod
    IMPORTANCE=0
    MAX_SIZE=2
    MIN_SIZE=2
    SERVER_CATEGORY=ora.hub.category
  SERVERPOOL
    NAME=ora.srvtest
    IMPORTANCE=0
    MAX_SIZE=2
    MIN_SIZE=2
    SERVER_CATEGORY=ora.hub.category
```

The line “NAME=Default” corresponds to the name of the policy in the policy set. The policy set itself is represented by the file “new_policy” created above.

Step 2: Update the current policy set

To update the current policy set, we just have to edit our policy set file “new_policy” and to add new policies above the Default one. We can copy-paste the DEFAULT policy and edit parameters.

As an example, imagine we want to allocate one server more for production during the nights and to allocate all servers to the production during week ends, to disable the development instances. We append the following policies after the Default one:

```
POLICY
  NAME=Nights
  SERVERPOOL
    NAME=ora.srvprod
    IMPORTANCE=0
    MAX_SIZE=3
    MIN_SIZE=3
    SERVER_CATEGORY=ora.hub.category
  SERVERPOOL
    NAME=ora.srvtest
    IMPORTANCE=0
    MAX_SIZE=1
    MIN_SIZE=1
    SERVER_CATEGORY=ora.hub.category
POLICY
  NAME=WeekEnds
  SERVERPOOL
    NAME=ora.srvprod
    IMPORTANCE=0
    MAX_SIZE=4
    MIN_SIZE=4
    SERVER_CATEGORY=ora.hub.category
  SERVERPOOL
    NAME=ora.srvtest
    IMPORTANCE=0
    MAX_SIZE=0
    MIN_SIZE=0
    SERVER_CATEGORY=ora.hub.category
```

Once the file is edited, it must be loaded into the policy set configuration using crsctl:

```
[grid@node01 ~]$ crsctl modify policyset -file /u00/app/oracle/admin/policies/new_policy
```

Step 3: Enable a new policy

Our policy set now contains three policies (Default, Nights and WeekEnds). The currently enabled policy is “Default":

```
[grid@node01 ~]$ crsctl status policyset | grep -i last
LAST_ACTIVATED_POLICY=Default
```

We are going to enable the “Nights” policy and see what happens. To enable a policy, use the following crsctl command as grid:

```
[grid@node01 ~]$ crsctl modify policyset -attr “LAST_ACTIVATED_POLICY=Nights"
CRS-2673: Attempting to stop 'ora.db2.db' on 'node02'
CRS-2677: Stop of 'ora.db2.db' on 'node02' succeeded
CRS-2672: Attempting to start 'ora.db1.db' on 'node02'
CRS-2676: Start of 'ora.db1.db' on 'node02' succeeded
```

In the previous output, we can see on node02 that Oracle has stopped DB2 and started DB1 instead.

We can confirm that srvprod has now three servers running using srvctl:

```
[grid@node01 ~]$ srvctl status srvpool
Server pool name: Free
Active servers count: 0
Server pool name: Generic
Active servers count: 0
Server pool name: srvprod
Active servers count: 3
Server pool name: srvtest
Active servers count: 1

[grid@node01 ~]$ srvctl status database -d DB1
Instance DB1_1 is running on node node01
Instance DB1_2 is running on node node04
Instance DB1_3 is running on node node02

[grid@node01 ~]$ srvctl status database -d DB2
Instance DB2_2 is running on node node03
```

Now we enable the “WeekEnds” policy. The remaing development server will be allocated to the production server pool and 4 production instances will be running:

```
[grid@node01 ~]$ crsctl modify policyset -attr “LAST_ACTIVATED_POLICY=WeekEnds"
CRS-2673: Attempting to stop 'ora.db2.db' on 'node03'
CRS-2677: Stop of 'ora.db2.db' on 'node03' succeeded
CRS-2672: Attempting to start 'ora.DB1.db' on 'node03'
CRS-2676: Start of 'ora.DB1.db' on 'node03' succeeded
```

All nodes are now part of the production pool:

```
[grid@node01 ~]$ srvctl status srvpool
Server pool name: Free
Active servers count: 0
Server pool name: Generic
Active servers count: 0
Server pool name: srvprod
Active servers count: 4
Server pool name: srvtest
Active servers count: 0
```

Four production instances are opened:

```
[grid@node01 ~]$ srvctl status database -d DB1
Instance DB1_1 is running on node node01
Instance DB1_2 is running on node node04
Instance DB1_3 is running on node node02
Instance DB1_4 is running on node node03
```

And the service for development is completely down:

```
[grid@node01 ~]$ srvctl status database -d DB2
Database is not running.
```

To reset to the initial policy, just use the same command with the “Default” policy attribute:

```
[grid@node01 ~]$ crsctl modify policyset -attr “LAST_ACTIVATED_POLICY=Default"
```

Conclusion

Policies-set represents an easy way to configure a whole policy managed cluster at will. We have seen how to edit a policy set using a text file generated using crsctl, but Oracle also documents how to manually define a policy set with command line utility, or using graphical tool. In my opinion, the text file is a better choice since it is very quick to edit a property in a text file.

However, we can regret that Oracle has not provided a built-in tool to schedule the policies. It seems that the only choices available to enable a given policy is to manually run crsctl, or to use a kind of scheduler (crontab or advanced task scheduler). ■

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All you need to know about Oracle Database licensing with VMware

You are running Oracle on VMware and are not 100% sure that you are compliant with the Oracle licensing rules? This article is for you! As I have been confronted several times with Oracle License reviews with VMware installations, I decided to write this article in order to provide answers to the most frequent questions I have received from customers.

Introduction

Licensing is not an easy topic to address and this difficulty is not specific to Oracle. Additionally, this difficulty increases with the number of different software editors and the related software policy you have to manage as a License Manager, IT manager or CTO. In this article, I will focus on Oracle licensing and more specifically on a very frequent use case: Oracle Database installed on a VMware ESX infrastructure. Indeed, there is really few information about this topic in Oracle's documentation and a simple mistake in your setup can easily cost you up to a million dollars. Lots of the information provided in this article originates from the Oracle Software Investment Guide[1]. However, the very first element to read is the Terms and Conditions specified in your Oracle contracts.

Oracle on VMware: supported, but not certified

Before diving into the Oracle licensing questions, the very first thing you have to know about any Oracle product, not only the Oracle Database, is that there is no Oracle product certified on a VMware architecture. However, all Oracle products are supported on VMware infrastructure. What does this mean, supported, but not certified? Oracle does not guarantee the proper working of these products on VMware architecture, but will support you in case of incidents not related to the VMware infrastructure. In cases where the VMware could be involved, the Oracle support may ask you to reproduce the incident on a certified environment. Below, you will find an extract of the Oracle Support Doc ID 249212.1.:

Oracle Support will assist customers running Oracle products on VMware in the following manner: Oracle will only provide support for issues that either are known to occur on the native OS, or can be demonstrated not to be as a result of running on VMware.

If a problem is a known Oracle issue, Oracle support will recommend the appropriate solution on the native OS. If that solution does not work in the VMware virtualized environment, the customer will be referred to VMware for support. When the customer can demonstrate that the Oracle solution

does not work when running on the native OS, Oracle will resume support, including logging a bug with Oracle Development for investigation if required.

If the problem is determined not to be a known Oracle issue, we will refer the customer to VMware for support. When the customer can demonstrate that the issue occurs when running on the native OS, Oracle will resume support, including logging a bug with Oracle Development for investigation if required[2].

[1] Oracle (March 2014), Oracle Software investment Guide
[2] My Oracle Support (2014), Support Position for Oracle Products Running on VMWare Virtualized Environments

Crash course on Oracle licensing terminology

Since explaining the subtleties of Oracle licensing without sharing some insight about the Oracle licensing terminology is very difficult, I will give you some explanation about the most common and basic Oracle licensing terms.

Oracle products are freely available for download on the Oracle website. However, before using it, you need to agree with the Oracle Technology Network (OTN) Developer License Terms. You can find an extract of the LICENSE RIGHTS Chapter below:

We grant you a nonexclusive, nontransferable limited license to use the programs only for the purpose of developing, testing, prototyping and demonstrating your application and not for any other purpose. ***If you use the application you develop under this license for any internal data processing or for any commercial or production purposes, or you want to use the programs for any purpose other than as permitted under this agreement, you must obtain a production release version of the program by contacting us or an Oracle reseller to obtain the appropriate license.*** *You acknowledge that we may not produce a production release version of the program and any development efforts undertaken by you are at your own risk.* ***We may audit your use of the programs***[3] ***.***

A license according to Oracle definition is **"the non-exclusive and limited right to use Oracle software under the agreed terms and conditions"**[4]. The overarching license rights are described in the Oracle Master Agreement (OMA), and the rights regarding specific products and services are described in the Ordering Document. Specifically, the OMA is the agreement, that details the standard rights granted, ownership, restrictions, warranties, disclaimers, confidentialities etc. The Ordering Document describes the specific products, types of licenses, number of users, level of support and discounts (if applicable) a customer has ordered and will receive.

In order to quantify and measure the way customers are using Oracle software, Oracle has introduced the notion of License Metric. According to Oracle the **"License Metric determines how the usage is being measured, when Oracle licenses its software to a customer"**[5]. For the technology related products (i.e. Oracle Database, Application Server products, Business Intelligence Technology products, Identity Management Products etc.) Oracle uses two metrics:

a. Processor
b. NUP (Named User Plus)

The **Processor License Metric** is based on the number of processor cores in the Enterprise Edition or Sockets in the Standard Edition and the Standard Edition One, in the server where the Database or Middleware product is installed and/or running. This metric is used where users are uncountable (i.e. website) or when the number of users changes frequently. It is important to notice that all processors have to be licensed even if one of them is deactivated in the BIOS. In the case of the Enterprise Edition, customers have to count the number of cores multiplied by the core factor in order to determine the number of Oracle processor cores. The core factor determines the coefficient used depending on the processor type. This core factor is defined in the core factor table[6] provided by Oracle at the following address:

http://www.oracle.com/us/corporate/contracts/processor-core-factor-table-070634.pdf

Vendor and Processor	Core Processor Licensing Factor
Sun and Fujitsu UltraSPARC T1 processor (1.0 or 1.2 GHz) Only named servers including: Sun Fire T1000 Server, SPARC Enterprise T1000 Server*, with 6 or 8-core 1.0 GHz UltraSPARC T1 processor Sun Fire T2000 Server, SPARC Enterprise T2000 Server*, with 4, 6, or 8-core 1.0 GHz, or 8 core 1.2 GHz UltraSPARC T1 processor	0.25
Sun Netra T2000, 1.0 or 1.2 GHz UltraSPARC T1 processor	0.25
S PARC T3 processor	0.25
Sun and Fujitsu UltraSPARC T1 1.4 GHz Only named servers including: Sun Fire T2000 Server and SPARC Enterprise T2000 Server*, with 8-core, 1.4 GHz UltraSPARC T1 processor	0.5
A MD Opteron Models 13XX, 23XX, 24XX, 32XX, 41XX, 42XX, 43XX, 61XX, 62XX, 63XX, 83XX, 84XX or earlier Multicore chips	0.5
Intel Xeon Series 56XX, Series 65XX, Series 75XX, Series E7-28XX, E7-28XX v2, Series E7-48XX, E7-48XX v2, Series E7-88XX, E7-88XX v2, Series E5-24XX, Series E5-26XX, E5-26XX v2, Series E5-46XX, E5-46XX v2, Series E5-16XX, Series E3-12XX or earlier Multicore chips	0.5
Intel Itanium Series 93XX or earlier Multicore chips (For servers purchased prior to Dec 1st, 2010)	0.5
Intel or AMD Desktop, Laptop/Notebook, or Netbook Multicore chips	0.5
Sun UltraSPARC T2+	0.5
SPARC64 VII+	0.5
SPARC64 X, SPARC64 X+	0.5
SPARC T4 processor	0.5
SPARC T5	0.5
SPARC M5	0.5
SPARC M6	0.5
Sun and Fujitsu SPARC64 VI, VII	0.75
Sun UltraSPARC IV, IV+, or earlier Multicore chips	0.75
Sun UltraSPARC T2	0.75
HP PA-RISC	0.75
IBM POWER5+ or earlier Multicore chips	0.75
All Single Core Chips	1.0
Intel Itanium Series 93XX (For servers purchased on or after Dec 1st, 2010) Intel Itanium Series 95XX	1.0
IBM POWER6	1.0
IBM POWER7, IBM POWER7+	1.0
IBM POWER8	1.0
IBM System z (z10 and earlier)	1.0
All Other Multicore chips	1.0
** SPARC Enterprise T1000 and SPARC Enterprise T2000 Servers may be sold and branded by Oracle, Sun Microsystems, Fujitsu or Fujitsu Siemens*	

Let's take a simple example in order to demonstrate the calculation method. An Oracle customer runs several Oracle Enterprise Edition Databases on one server equipped with two sockets. Each socket is equipped with an Intel XEON X7560 8 cores processor. The usual Intel core factor is 0.5 (c.f. core factor table). The customer has to license 2 sockets x 8 cores x 0.5 core factor = 8 Oracle processors. If this customer is running the same databases on a Standard Edition or a Standard Edition One, he has to license two Oracle processors since two sockets are occupied.

The **NUP Metric** is based on users and non-human operated devices accessing the database. This metric is usually used when users can be counted. Customers using this metrics have to license either the number of NUP accessing the Oracle software or the minimum NUP – whichever NUP is higher. The calculation rule for the minimum NUP customers have to license is 25 NUP per Oracle Processor for the Enterprise Edition and 5 NUP for the Standard Edition and the Standard Edition One. Please note that the number of NUP for the Standard Edition and the Standard Edition One is only 5, since the minima in the Standard Edition and the Standard Edition One is not related to the number of processor license.

Let's come back to the example in the processor section: For the same architecture, the minimum Named User Plus this customer has to license according to the minima rules would be 8 Oracle Processor x 25 = 200 NUP on an Oracle Enterprise Edition and 5 on an Oracle Standard Edition and Standard Edition One.

The same rules described above regarding the Enterprise Edition apply for the Enterprise Edition Options.

Differences between the different Oracle Database Editions

The main differences between the different Oracle Database Editions are:

- Oracle Database Enterprise Edition has no license restrictions. This is the only edition that allows licensing Database options (e.g. Partitioning, OLAP, Data Mining, Spatial, Enterprise Management Pack etc.). In addition, Oracle Database Enterprise includes an additional set of features that are not present in other editions (e.g. Data Guard, Transportable Tablespace, Materialized View Query Rewrite, Cross-Platform Backup, Flashback Table, Database, Transaction Query etc.). In addition this edition is the only one allowed to be monitored by Grid Control or Cloud Control.

- Oracle Database Standard Edition can only be licensed on servers that have a maximum capacity of 4 sockets. This edition encompasses Real Application Cluster, however, the addition of sockets into the cluster nodes cannot be superior to 4. The maximum number of CPUs defined is meant for the entire Real Application Cluster infrastructure; it is not a per node maximum.

- Oracle Database Standard Edition One may only be licensed on servers that have a maximum capacity of 2 sockets. Unlike the Standard Edition, the Standard Edition One does not include Real Application Cluster. In addition, unlike the Standard Edition and the Enterprise Edition, the Standard Edition One does not include Automatic Workload Management.

- Oracle Database Express Edition (XE) can be installed on any size of host machine with any number of CPUs (one database per machine), but XE will store up to 11GB of user data, uses up to 1GB of memory and one CPU on the host machine. Many features integrated in other edition are not present in the Express Edition.

An exhaustive list of differences is provided on the following URL:
www.oracle.com/us/products/database/enterprise-edition/comparisons/index.html

[3] Oracle (2014), Oracle Technology Network Developer License Terms
[4] Oracle (2013), Oracle License and Service Agreement
[5] Oracle (March 2014), Oracle Software investment Guide
[6] Oracle(2014), Oracle Processor Core Factor Table

Hard and soft server partitioning

According to Oracle's definition: **"Partitioning occurs, when the CPUs on a server are separated into individual sections, where each section acts as a separate system. Sometimes this is called "segmenting."There are several hardware and software virtualization technologies available that deliver partitioning capabilities, with varying degree of resource allocation flexibility."** [7]

A partitioning strategy is more and more frequently used by customers. The underlying technologies and products aim to achieve several goals such as:

- Workload balancing by allocating more or less CPU power to specific systems
- Architecture consolidation by running multiple and different operating systems or multiple versions of the same operating system on the same physical server in order to optimize the resource usage
- The deployment of economic models such as "Pay-As-You-Grow" and "Capacity on Demand"

Oracle distinguishes two main types of partitioning:

Soft partitioning

Soft partitioning is defined by Oracle as a flexible way to segment the operating system using OS resource managers, since the CPU capacity allocated to the operating system running an Oracle database can be changed fairly easily. Examples of such partitioning types include (non-exhaustive list): Solaris 9 Resource Containers, AIX workload Manager, Oracle VM and VMware.

The key message regarding soft partitioning is: **"soft partitioning is not permitted as a means to determine or limit the number of software licenses required for any given server."**[8]

In other terms, customers using such soft partitioning technologies have to license the totality of cores or sockets of the physical systems.

Hard partitioning

Hard partitioning is defined by Oracle as "a way to physically segment a server, by taking a single large server and separating it into distinct smaller systems. Each separated system acts as a physically independent, self-contained server, typically with its own CPUs, operating system, separate boot area, memory, input/output subsystem and network resources."[9]

Oracle approved hard partitioning technologies as listed in this section of the policy document. They are permitted as a means to limit the number of software licenses required for any given server or a cluster of servers. Approved hard partitioning technologies include: Dynamic System Domains (DSD) -- enabled by Dynamic Reconfiguration (DR), Solaris Zones (also known as Solaris Containers, capped Zones/ Containers only), LPAR (adds DLPAR with AIX 5.2), Micro-Partitions (capped partitions only),vPar, nPar, Integrity Virtual Machine (capped partitions only), Secure Resource Partitions (capped partitions only), Fujitsu's PPAR.

Oracle VM Server can be used as a hard partitioning technology if customer bind vCPUs to physical CPU threads or cores as explained in the following paper: **http://www.oracle.com/technetwork/server-storage/vm/ovm-hardpart-168217.pdf**

Licensing of VMware on Oracle.

As described in the soft partitioning chapter, VMware is considered by Oracle as a soft partitioning technology.

Some white papers provided by VMware such as the one named "Understanding Oracle Certification, Support and Licensing for VMware Environments"[10] state that it is possible to partially license cluster using DRS Host Affinity.

This document has never been approved by Oracle and you won't find anything about such statements in the Oracle documentation, the Oracle licensing terms, or in the Oracle support. Oracle does not allow to partially license an ESX cluster using DRS technology. Up to VMware 5.0, customers running a virtual machine with Oracle installed had to fully license the physical hosts composing the VMware cluster. Since VMware 5.1, if only one Virtual Machine is running Oracle managed within a vCenter Server Instance, you have to fully license all the physical hosts managed by this vCenter Server instance and not only the cluster instances. This decision has probably been taken by Oracle due to the new features coming with VMware vSphere 5.1, particularly one named vSphere vMotion: This new feature leverages the advantages of vMotion (zero-downtime migration) without the need for shared storage configurations. Indeed, this new vMotion capability applies to the entire network.

In addition, in order to license only the vCenter Server Instance (or the cluster up to VMware 5.0) where Oracle products are running, you have to dedicate a separate physical storage to this vCenter (or cluster up to VMware 5.0). Virtual storage separations such as Vplex technology or dedicated LUN are not considered as a physical separation.

Regarding Oracle Standard Edition, it can be licensed if each individual physical server on the vCenter Server instance does not exceed the maximum capacity of 4 sockets. Oracle Standard Edition One can also be licensed if the maximum capacity of each individual physical server in the vCenter Server Instance does not exceed 2 sockets.

[7,8,9] Oracle (2013), Oracle Partitioning Policy

[10] VMware (2014), Understanding Oracle Certification, Support and Licensing for VMware Environments

Practical case

In the example illustrated in Figure 1 a customer has a VMware infrastructure of version 5.1 with two servers in a cluster and a third separated server. All servers are managed by the same Vcenter Server Instance.

- The servers in the cluster are equipped with two Intel Xeon with 8 cores each. The third server is equipped with one Intel Xeon with 4 cores.
- Each server on the cluster hosts two Virtual Machines. 8 cores are allocated to each Virtual Machine. The third server has only one Virtual Machine with 4 dedicated cores.
- Oracle Database Enterprise Edition is installed on one Virtual Machine on the cluster (VM1). This VM can only run on server1. The database is used by 300 users.

In this example, the following restrictions apply:

- VMware is counted as soft partitioning and therefore any dedication or restriction of the Oracle programs using it, is considered as soft partitioning, and hence you will still need to license the whole environment.
- Since no physical server has more than two sockets, Oracle Standard Edition or the Enterprise Edition One can be licensed.
- Oracle will only provide support for issues that either are known to occur on the native OS or can be proven not to be a result of running on VMware.

The customer will have to purchase:

- 450 (36 cores x 0.5 core factor x 25) named User Plus in Enterprise Edition in order to respect the minima rules.
- Alternatively, the customer can acquire 18 (36 cores X 0.5 Intel core factor) Oracle Processor of Oracle Enterprise Edition.

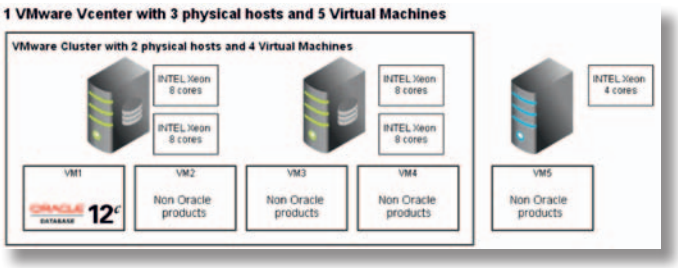

Figure 1: Customer VMware 5.1 architecture

Architecture solutions

Taking into consideration the facts described in the preceding section, what are the possible solutions in order to be compliant with Oracle licensing terms and conditions? Below, you can find a non-exhaustive list of solutions. I voluntary decided to mostly focus on Oracle propositions:

1. **Dedicating and licensing a new VMware vCenter Server Instance to Oracle products** with a separated physical storage for Oracle in order to limit the Oracle license footprint to this specific vCenter Server Instance. This solution can be interesting if your company strategy is to install VMware systematically as an "under layer" for each system. The cost of this solution mostly depends on the chosen storage system.

2. **Consolidation on a physical host.** It is indeed perfectly possible to run several Oracle products of different version on the same physical host. Of course, this solution is not the most flexible, but will cost you close to nothing in terms of license. However, it will be mandatory to migrate all your Oracle products running on Virtual Machine to a physical server and this can be time consuming and tedious. The price of this solution mainly depends on the chosen physical server.

3. Using **a hard partitioning technology** such as the one presented in the chapter "hard partitioning". The price and time needed to migrate from VMware to another system varies greatly from one system to another depending on the chosen hard partitioning technology. For instance, Oracle VM (in hard partitioning mode) is capable of loading both VMware and Hyper-V Virtual Machines and converting them automatically to an Oracle VM. An Oracle VM with a one year support will cost you 599 dollars per server with a maximum of 2 CPUs or 1'199 dollars per server with any number of CPUs[11]. This solution also offers the advantage to rapidly deploy the application via the Oracle VM preconfigured templates.

4. **Switching on an Oracle Appliance** such as the Oracle Database Appliance (ODA): Indeed, this appliance can be installed in a bare metal or in a virtualized mode. One of the advantages of this technology is that you can license a minimum of 2 cores per node (a total of 4 cores) in a virtualized mode and you can activate up to 48 cores (with increment by 2)! Nowadays, finding a solution of two servers with only 4 overall cores, meaning two Oracle Processor Licenses, is nearly impossible. Unfortunately, one drawback is that you cannot license

[11] Oracle (June 2014), Oracle Linux Support and Oracle VM Support Global Price List

the Standard Edition on this box. A license revalorization is however frequently possible, which means you do not have to pay the full conversion price from the Standard Edition to the Enterprise Edition. Finally, Oracle Database Appliance will offer you a simple and highly available solution. The price of this appliance according to Oracle's Engineered Systems price list is 60'000 dollars[12].

Figure 2: Oracle Database Appliance X4-2

5. **Using the Server Trusting Partitioning mode.** Oracle provides a relatively new third server partitioning mode. It allows the use of Oracle VM Server (OVM) as a means to limit the number of Oracle Processor licenses required in order to license a sub-capacity of total physical cores without the drawback of OVM in hard partitioning mode. This server partitioning mode is named: Trusted partitioning. The usage conditions of trusted server partitioning are the following:

Figure 3: Oracle Virtual Compute Appliance

   a. Using an approved Engineered System (such as Exalogic Elastic Cloud, Exalytics In-Memory Machine, or Oracle Virtual Compute Appliance – OVCA)
   b. Using an Oracle Enterprise Manager 12.1.0.2 or higher (either in connected or disconnected mode) to monitor VMs participating in Oracle Trusted Partitions.

Customers will get the following advantages when running OVCA with trusted partitioning:

- You do not have to license the totality of cores: Only used physical cores have to be licensed.
- On an OVCA, two Virtual CPUs (vCPU) are counted as equivalent to a physical core. For instance, licensing 4 Oracle processors enables you to use 8 physical cores and 16 vCPUs.
- Licenses are procured in increments of 2 physical cores (1 Oracle Processor), which provide a high level of flexibility.
- On an OVCA, you can run a Standard Edition, which is not possible on an ODA. Thanks to Oracle Virtualization, you can even run other RDBMS, such as Microsoft SQL Server on Windows, for instance.
- Even if this appliance is an important investment in terms of capital expenditure, such a solution could drastically reduce your operational expenditures.

[12] Oracle (2014), Oracle Engineered System Price List

Conclusion

Running Oracle products on VMware offers flexibility advantages. However, if you do not have a deep knowledge of Oracle licensing rules, it can expose you to high financial risks. In addition, the probability of facing a license control is quite high: In French-speaking Switzerland alone, there are more than 20 Oracle LMS (License Management Services) audits each year.

The core message of this article is not to switch from VMware to another solution. However, if you have any uncertainty, you should ask an Oracle LMS or Oracle sales specialist before designing your Oracle architecture. Of course, dbi services can also help you maintaining your Oracle licenses either through our SLA or through periodic reviews.

I hope that this article has helped you getting a better insight of Oracle licensing on VMware. ■

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SMS >>>

DOAG 2015 BI, Exaday und Datenbank: Call for Papers eröffnet

Gleich für drei Konferenzen der DOAG können Referenten ab sofort ihre Vortragsthemen vorschlagen. Gesucht sind klassische Beiträge von 45 Minuten. Der Eintritt zur jeweiligen Konferenz ist für Referenten kostenfrei. Here we go:

- **DOAG 2015 Business Intelligence** – «In-Memory», «Big Data & BI», «Performance», «Modellierung» und «Grundlagen» sind die Themen der DOAG 2015 Business Intelligence. Bis zum 20. Januar 2015 können Referenten ihr Vortragsthema vorschlagen.
- **DOAG 2015 Datenbank** – Ebenso bis zum 20. Januar 2015 läuft der Call for Papers für die DOAG 2015 Datenbank. Die Datenbank Community der DOAG sucht nach spannenden Vorträgen aus den Themenbereichen «Administration (Backup & Recovery, Monitoring, Tuning)», «Security», «Migrationen/Upgrades», «Hochverfügbarkeit/Desaster Recovery», «Optionen der Datenbank: In-Memory, Multitenant DB», «Replikationstechniken» und «Database as a Service».
- **DOAG 2015 Exaday** – Wer sein Know-how zu einem der Exadata-Themenschwerpunkte «Lifecyle», «Features» oder «Produkt» weitergeben möchte, kann bis zum 22. Januar 2015 einen Vortragsvorschlag einreichen. Exadata- und ODA-Administratoren sowie Entscheidungsträger und Systemarchitekten erfahren auf der DOAG 2015 Exaday alles Wichtige rund um die Fragen «Wie funktionieren die integrierten Systeme im Einsatz?» und «Was muss für die Evaluierung von integrierten Systemen berücksichtigt werden?». Die Veranstaltung findet am 28. April 2015 in Frankfurt statt.

Mehr Infos: http://tinyurl.com/nuv5gfe

Alexandre Tacchini, Benjamin Gaillard, Fabien Grange, Synchrotech

De la données agrégées à la données atomiques: Quelles solutions sont envisageables avec la suite OBIF?

1 Introduction

De nos jours, un grand nombre d'entreprises désire analyser ses données à tous les niveaux; du plus agrégé au plus atomique pour répondre, au travers d'une même plateforme, aux différents besoins de reporting d'entreprise, de tableaux de bord, de requêtes à la demande ou d'indicateurs stratégiques de pilotage (scorecarding).

Au travers la mise en œuvre des différentes solutions de BI intégrées fournit par la plateforme « Oracle Business Intelligence Foundation Suite (OBIFS) », les clients améliorent leur visibilité du business, leur alignement sur la stratégie et leur réactivité face au marché.

1.1 De quoi parle-t-on ?

Au travers de cet article, Synchrotech vous présente deux concepts d'implémentation de la suite OBIF qui permettent aux utilisateurs finaux de naviguer entre leurs données détaillées et agrégées, dans un outil de restitution unique. Les deux modules (Essbase et OBIEE) permettant la mise en place de ces deux concepts répondent à un besoin identique tout en étant complètement différents et complémentaires.

1.1.1 Le module Essbase

Les différents modes d'agrégation et de stockage des cubes OLAP Essbase (ASO - Aggregate Storage Outline et BSO – Block Storage Outline) permettent de combiner la puissance de calcul et de simulation des applications BSO à la force d'agrégation des applications ASO au travers des partitions qui lient les applications. Ces différents liens permettent ainsi aux utilisateurs finaux de se connecter à un seul outil de restitution pour voir les différents jeux de données.

1.1.2 Le module OBIEE

Grace au multi-sourcing, OBIEE permet également de fournir au travers d'une interface de restitution unique un accès à différentes sources agrégées ou détaillées. Les différentes couches du Repository OBIEE permettent dans un premier temps la connexion à de multiples sources de données (OLAP, OLTP, CSV, XML, …), dans un deuxième temps l'intégration des ses sources entre elles et dans un dernier temps la préparation des données intégrées dans un rendu utilisateur compréhensible pour les utilisateurs finaux.

2 Utilisation d'Essbase

2.1 Description

La mise à disposition de données analytiques à travers Essbase donne une grande flexibilité de navigation et d'analyse au travers de SmartView ou de tout autre outil de Reporting tel qu'OBIEE.

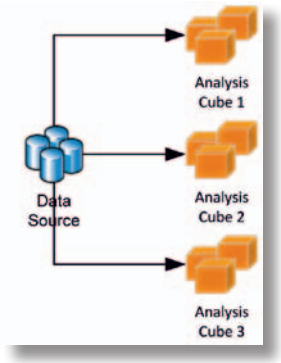
La haute granularité des données peut engendrer différents problèmes dans le traitement de l'information avec Essbase. Les problèmes peuvent survenir autant du côté des utilisateurs finaux que des administrateurs. Les problèmes rencontrés fréquemment sont les suivants :

- Trop d'information disponible peut masquer l'information pertinente et rendre l'analyse de ladite donnée plus difficile.
- La navigation dans les données peut être difficile lorsqu'un grand nombre de dimensions est disponible
- La grande masse de données rend difficile la gestion des structures et la mise à jour des cubes Essbase.

Au travers de cet article, nous allons décrire un exemple d'architecture possible avec Essbase qui permet à l'utilisateur de disposer de l'ensemble des données, sans pour autant avoir à naviguer en tout temps au sein d'un modèle dimensionnel gigantesque. Cette architecture permet aux administrateurs d'avoir des gains de performance notables lors des chargements de données et des changements de structure.

2.2 Principe

Scinder les données en différents DataMart est un principe connu et répandu dans la mise en place d'entrepôts de données. Cette pratique permet d'augmenter les performances du moteur relationnel en évitant de requêter sur des données inutiles lors ceci n'est pas requis. Ce même principe peut être appliqué à Essbase en n'hésitant pas à créer différents cubes selon les thèmes analytiques désirés par le business. Ceci même si ces cubes utilisent des données identiques.



La figure ci-contre démontre l'utilisation de plusieurs cubes étant alimentés par une source unique. Cette approche est nécessaire lorsque les règles de calculs, d'agrégation ou le détail des données sont différents selon les besoins. Nous appellerons ces cubes d'analyse « cube de Reporting ».

La particularité de ce modèle est la réplication potentielle d'une même donnée dans différents cubes avec les éventuels problèmes que cela peut induire :

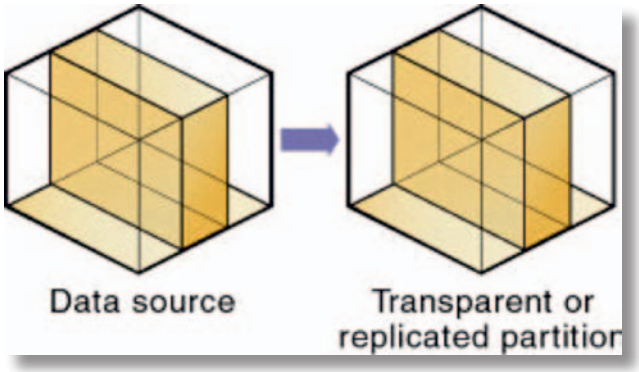
- Augmentation du volume de données (contraintes de stockage, de puissance serveur pour le calcul, …)
- Augmentation du risque d'erreur lors des travaux ETL

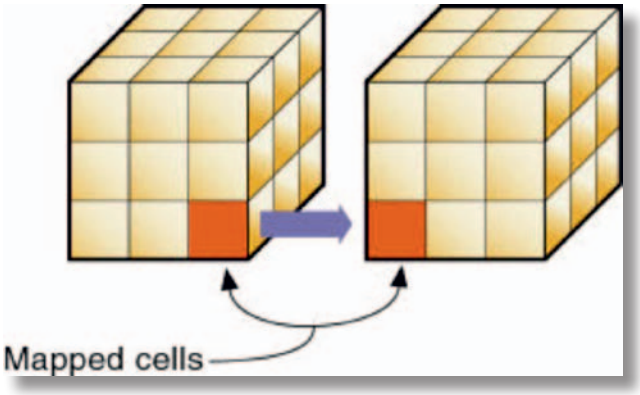
Tenant compte de ces risques et contraintes, Essbase propose des solutions de partitionnement qui permettent la réutilisation des données d'un cube à l'autre.

Le partitionnement est un mécanisme intégré à Essbase permettant de partager une région d'un cube avec un autre. De plus, une application partitionnée peut être transversale sur plusieurs serveurs.

Essbase met à disposition trois types de partitions, soit :

- **Répliquée**
  Une copie des données est effectuée entre la base source et cible. La copie est entièrement gérée par Essbase, sauf son déclenchement ponctuel.
- **Transparente**
  La donnée n'est pas stockée du côté cible. Essbase effectue la requête de la cible vers la source pour récupérer la donnée et la transmettre à l'utilisateur.
- **Liée**
  Envoie l'utilisateur d'une cellule d'une base vers une autre cellule d'une autre base. La partition liée donne une autre perspective aux données.

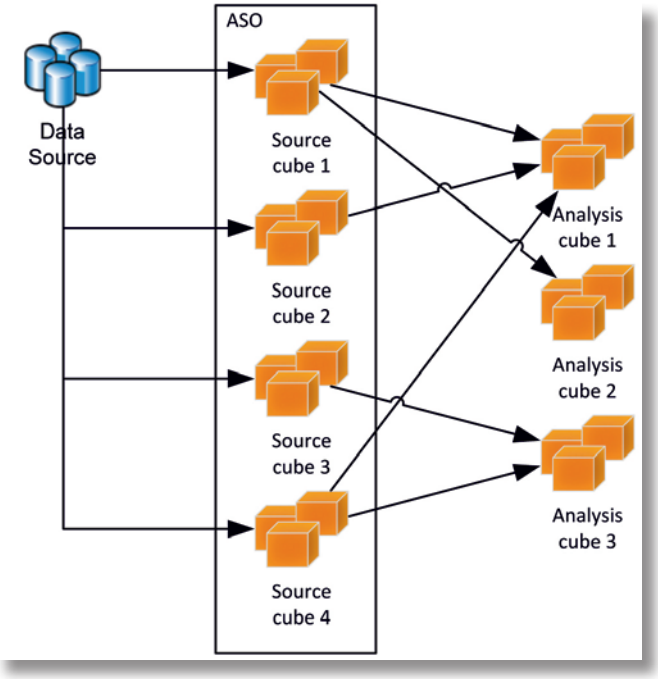




Mettant à profit les technologies de partitionnement ainsi que les deux modes de stockage fournis par Essbase (ASO / BSO)[1], l'architecture proposée ici va multiplier les cubes de Reporting afin de cibler parfaitement les domaines d'analyse tout en récupérant les données d'une multitude de cubes source afin de ne pas multiplier la données et ainsi profiter de la performance d'agrégation de l'ASO.

2.3 Architecture

L'architecture cible ressemblera à la figure ci-dessous.



L'usage du mode de stockage ASO pour les cubes source se justifie par le grand nombre de dimensions et de croisements que peuvent représenter ces bases de données. En effet, il est fort probable que beaucoup de détails soit inclus dans ces bases de données et rendrait du coup l'utilisation du BSO plus compliqué.

On peut donc constater que les travaux d'agrégation seront fortement délégués aux cubes sources, profitant ainsi de la puissance de calcul de l'ASO. Les formules et spécificités du Reporting seront alors réalisées dans les cubes de Reporting. Ceux-ci pouvant être en ASO ou BSO.

Cette architecture amène l'avantage certain de ne pas se préoccuper de la réutilisation des données dans les différents cubes, ceci étant géré par les partitions.

Les cubes de Reporting auront donc leur structure propre. L'objectif étant d'améliorer l'expérience utilisateur dans la navigation en préparant au mieux la structure de Reporting. Le détail désiré peut être variable. Ainsi, les cubes de Reporting peuvent prendre le parti de mettre à disposition uniquement un très haut niveau. Le niveau détaillé restant la fonction des cubes sources.

2.3.1 Design des cubes sources en ASO

Le design de la solution consiste tout d'abord à trouver la bonne maille afin de scinder les données en plusieurs cubes. On peut imaginer travailler sur plusieurs aspects : la période (un cube par année, mois), le type de données (ventes, profitabilité, …) ou tout autre élément différenciateur dans la donnée.

[1] Le mode Hybrid Essbase n'est pas couvert dans cet article, même s'il apporte indéniablement de nouvelles perspectives dans la manière d'organiser son architecture Essbase.

Une fois la maille choisie, l'identification des dimensions est importante et un flux unique et commun pour tous les cubes est nécessaire. Avoir des dimensions similaires entre tous les cubes sources et les cubes de Reporting va aider grandement à la mise en place des partitions (i.e. : une dimension représentant la structure des produits devra être la même dans tous les cubes).

La configuration spécifique des agrégations est similaire à tout autre cube ASO. La phase de «benchmarking» et de «fine tuning» est aussi nécessaire afin de s'approcher des performances optimales de consultation des données.

2.3.2 Design des cubes de Reporting en BSO

Les cubes de Reporting ne contiendront que peu, voir pas de données. Et ce surtout si les partitions utilisées sont de type transparent. On pourrait penser que la configuration de la densité des dimensions est alors inutile du moment qu'Essbase ne va pas stocker de données sur disque, et n'aura donc pas besoin d'organiser tout cela en blocks.

Tout cela est sans compter sur la manière dont Essbase travaille lorsqu'un utilisateur effectue une requête sur la base. Essbase va alors identifier quels sont les blocks de données nécessaires et va interroger la partition source en rapatriant l'entier des blocks. Une mauvaise organisation de la base peut engendrer des problèmes récurrents de performance lors de la consultation des données.

Il peut être également intéressant de transgresser les règles de configuration «standards» de la densité des dimensions. Prenons pour exemple un cube avec une dimension temps qui comporte un grand nombre de membre. Par définition, la dimension période est généralement denséw. L'entier des périodes sera alors compris à l'intérieur d'un block. Ce qui implique qu'à chaque requête, toutes les périodes seront rapatriées à travers la partition (et donc à travers le réseau). Ceci est peut-être inutile car la majorité des utilisateurs concentrent leur Reporting sur les périodes actuelles (mois courant, mois précédent, …). Envisager cette dimension en sparse peut être une piste d'amélioration des performances de consultation des données.

Il est par contre sûr que toute définition d'un cube virtuel usant des partitions transparentes comme source nécessite la même attention en configuration qu'un cube traditionnel. Et personne n'évite le passage par le «benchmarking».

2.3.3 Industrialisation

La séparation des données en une multitude de cube ajoute de la complexité aux administrateurs dans la gestion des structures et des chargements de données. En effet, au lieu de gérer les cubes de manière indépendante, il sera alors nécessaire de s'assurer que les Outlines soient parfaitement alignées lorsque celle-ci doivent évoluer.

L'ETL est ici particulièrement important. Le développement d'un flux spécifique associé à la surveillance de ces changements est requis. Il prend alors le rôle d'un chef d'orchestre.

Lors de la mise à jour des données dans les cubes, le chef d'orchestre a la responsabilité de contrôler minutieusement si les métadonnées ont changés. Dans un tel cas, il identifie alors les conséquences de ce changement. Certains changements peuvent être traités de manière simple, à l'instar d'un changement de description, de l'ajout d'un produit. Le chef d'orchestre se chargera alors de déclencher les jobs ad-hoc de chargement de la nouvelle structure, sans pour autant toucher aux données associées. L'avantage d'une telle architecture réside dans le traitement parallèle des travaux. Ainsi, tous les cubes verront leurs Outlines mises à jour. A noter également qu'Essbase propose une méthode de synchronisation des Outlines, mais celle-ci a ses limites surtout dans les cas où la partition est répliquée ou ne représente qu'une partie du cube et non toute la base.

Lors de changements plus conséquents (une règle de dérivation des données par exemple), le chef d'orchestre lancera alors des travaux de réalignement des données ainsi que le chargement dans tous les cubes concernés.

2.3.3.1 Mise à jour simultanée dans les cubes de Reporting

L'architecture améliore grandement la disponibilité des données. Dans le cas d'utilisation de la même donnée au sein de plusieurs cubes de Reporting (un chiffre d'affaires par exemple), celui-ci sera simultanément mis à jour dans tous les cubes de Reporting une fois le cube source chargé de la nouvelle donnée. Un gain indéniable pour éviter la redondance de l'information et la multiplication des processus de chargement avec les risques inhérents au traitement/stockage multiple de la même information.

2.4 Du côté de l'utilisateur

L'utilisateur final aura alors à disposition les cubes de Reporting. Ceux-ci sont clairement orientés pour améliorer l'expérience et limiter la phase d'apprentissage de l'utilisateur pour la navigation. L'usage intensif des partitions peut ralentir quelque peu la performance des requêtes, mais un «fine tuning» appliqué permettra de limiter cette impression.

Dans le besoin de plus de détail, celui-ci peut en tout temps se connecter également aux cubes sources et ainsi profiter d'un niveau de navigation plus étendu.

2.5 Conclusion

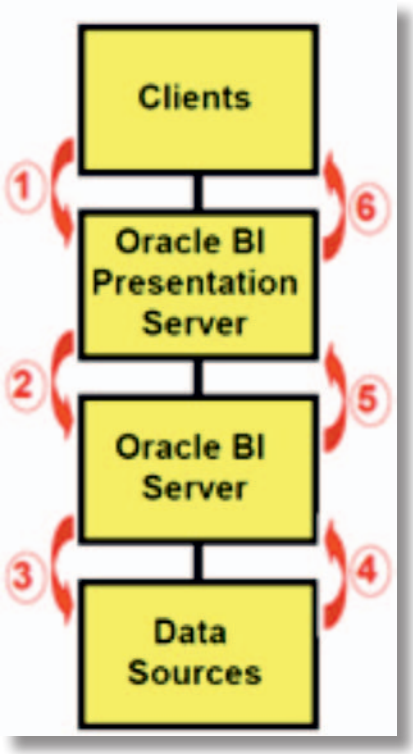
Ce type d'architecture réduit considérablement le stockage en faisant la promotion de la réutilisabilité des données. Néanmoins, lors de changement de structure, les processus lancés sur le serveur Essbase peuvent être nombreux et gourmands en ressources. Dans cette optique, l'utilisation de l'Appliance Exalytics se révèle être appropriée car le traitement en parallèle des chargements ainsi que la mise en mémoire de la données augmente de manière significative les performances. L'utilisation des 4TO de RAM et des 128 cœurs de la machine procure un gain non négligeable lorsque l'on doit recharger les données de plusieurs dizaines de cubes simultanément.

3 Utilisation d'OBIEE

3.1 Description

Une de ces solutions consiste à analyser les données depuis l'outil d'oracle «Oracle Business Intelligence Entreprise Edition».Ci-dessous, nous allons décrire comment fédérer des informations provenant de différentes sources. Cette fédération permettra d'effectuer une analyse à un niveau agrégé, mais aussi de pouvoir descendre à un niveau atomique sur un axe, soit horizontal, soit vertical.

3.2 Fonctionnement

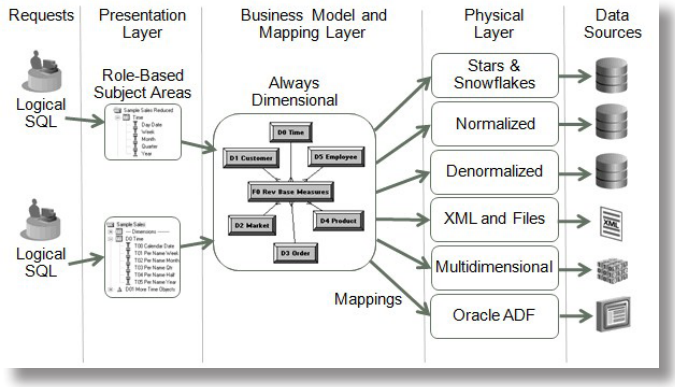


1) L'utilisateur soumet une requête via un Dashboard
2) Le BI Presentation crée et envoie une requête logique
3) Le BI Server optimise la requête logique pour créer la requête physique, puis l'envoie à la source de données.
4) La source de données envoie le résultat de la requête physique.
5) Le BI Server reçoit le jeu de données demandé et transforme ces données selon les règles établies pour l'envoyer au BI Presentation.
6) Le BI Presentation formate les données pour l'afficher dans le Dashboard.

3.3 Multi-source

Le Repository d'OBIEE est représenté par 3 différentes couches :

- La couche 1, dite physique (modèle de données depuis la base de données source).
- La couche 2, dite business model et mapping (définition d'un schéma en étoile avec les règles business).
- La couche 3, dite de présentation (définition des éléments visibles aux utilisateurs finaux)



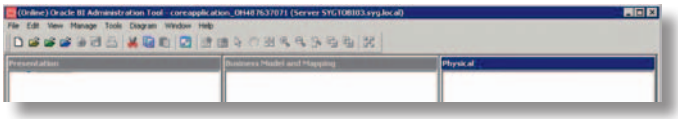
Source :
http://gerardnico.com/wiki/dat/obiee/logical_business_model

Le Repository, à défaut de son nom, stocke seulement les métadonnées du BI Server. Nous y retrouverons entreposés les informations suivantes :

- Sécurité des données
- Modèle de données
- Agrégation
- Cache
- Connections
- Variables de session et variables

OBIEE possède la faculté de lire plusieurs sources de données différentes dans sa première couche.

Chacune de ces sources peut être une source différente car OBIEE utilise plusieurs connecteurs (Oracle, Essbase, SQL server, Microsoft Access, fichiers plats, SAP, XML, …)



3.4 Fédération

La fédération consiste en l'intégration de 2 ou plusieurs sources de données dans un certain niveau de granularité joint par une ou plusieurs dimensions.

Selon le niveau de granularité, une distinction est faite pour cette fédération :

- Fédération horizontale
- Fédération verticale

3.4.1 Fédération horizontale

Nous parlerons de fédération horizontale lorsque l'on se trouvera au même niveau de granularité.

3.4.1.1 Cas pratique – Ajout d'une mesure

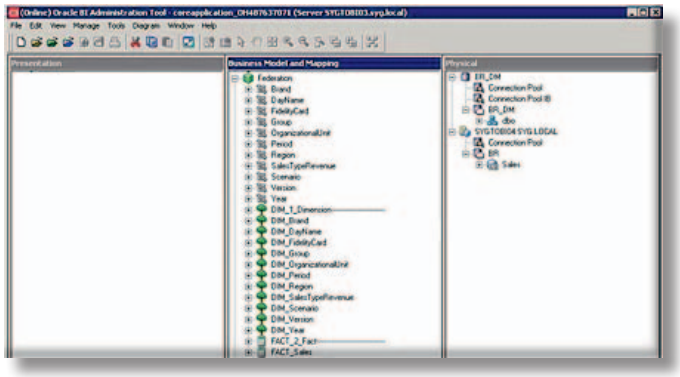
Nous disposons de 2 sources de données, un cube contenant des informations à un niveau agrégé et un Data-Mart contenant toutes les données détaillées.

Dans ce cas précis, nous désirons créer un rapport sur le chiffre d'affaires des points de vente et sur le chiffre d'affaires du 10 du mois.

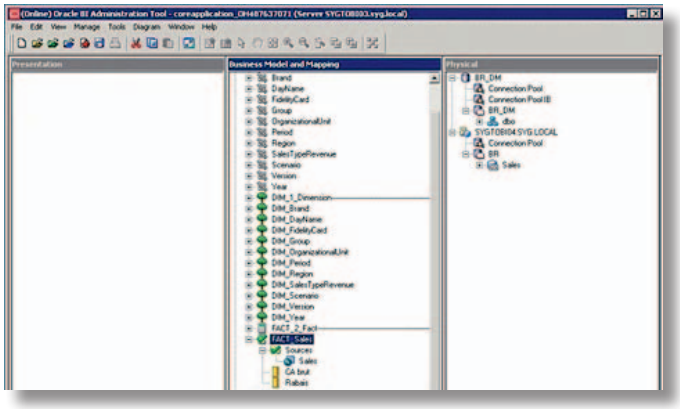
Après analyse, nous constatons que tous les champs sont disponibles dans notre cube sauf la clé d'indicateur de performance, la mesure du « 10 du mois ».

3.4.1.2 Exemple de mise en place

Sur la base des sources importées dans la couche physique, c'est dans la couche 2 que la fédération des métadonnées se fera.



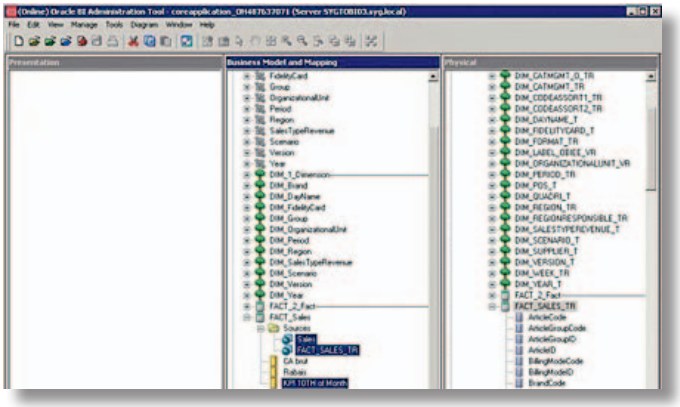
Un business model sera créé sur la base du cube pour disposer de toutes les métadonnées à un niveau agrégé. L'importation du cube permettra la création automatique du business model avec tous ses objets. Cela inclura les tables dimensionnelles, les tables de fait, les hiérarchies et les jointures.



Chaque dimension présente contiendra les métadonnées mises à plat avec un répertoire « Source ». Ce dernier contiendra toutes les informations d'où proviennent les données.

Par défaut, les mesures disponibles seront celles définies dans le cube. Elles seront stockées dans la table des faits. Les mesures dont nous désirons obtenir pour une analyse plus précise sont manquantes. Ces dernières sont seulement disponibles à travers une autre source de données.

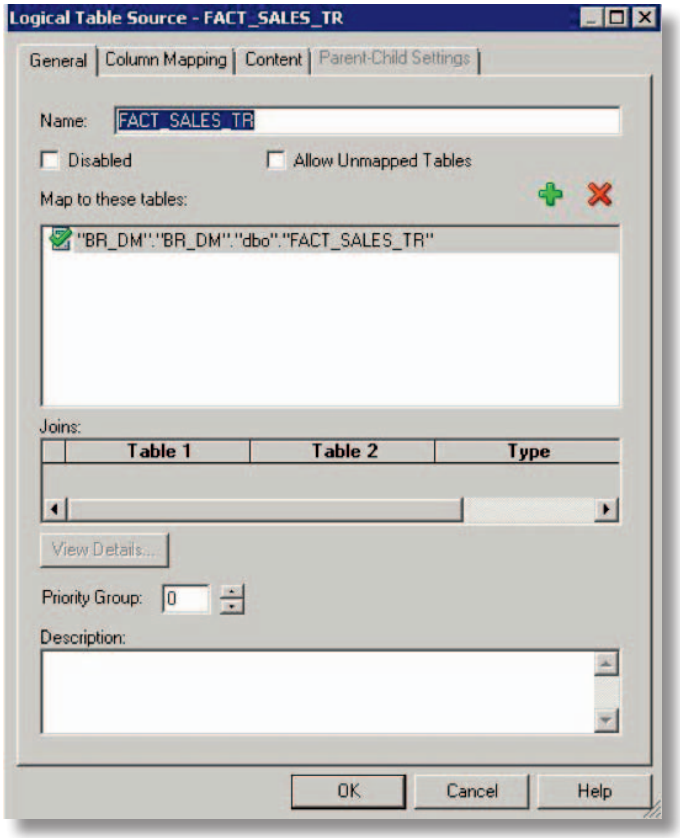
Il est très simple de manipuler les données d'une couche à une autre. Il suffit de faire un cliquer/glisser d'une fenêtre à l'autre.

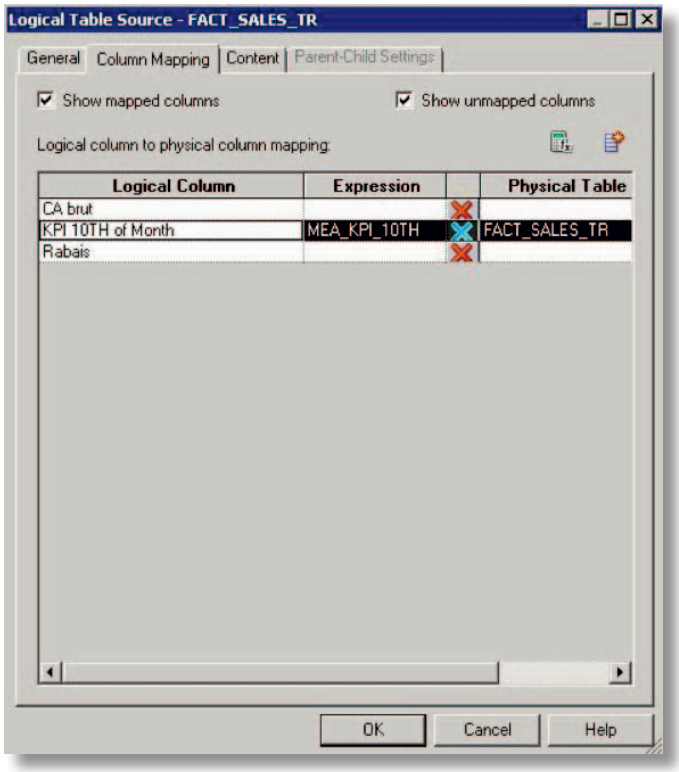


Lors de l'ajout de la mesure supplémentaire dans la table de faits logique, une nouvelle source s'ajoute dans le répertoire « Sources » et on y retrouve aussi notre nouvelle colonne.

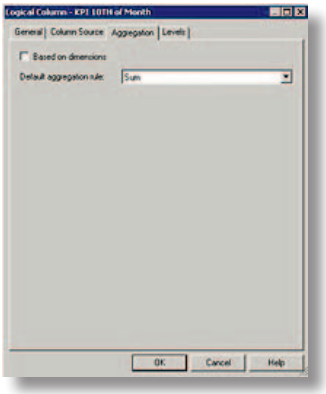
Ce répertoire « Sources » contiendra les informations du lien entre la couche physique et la couche business.

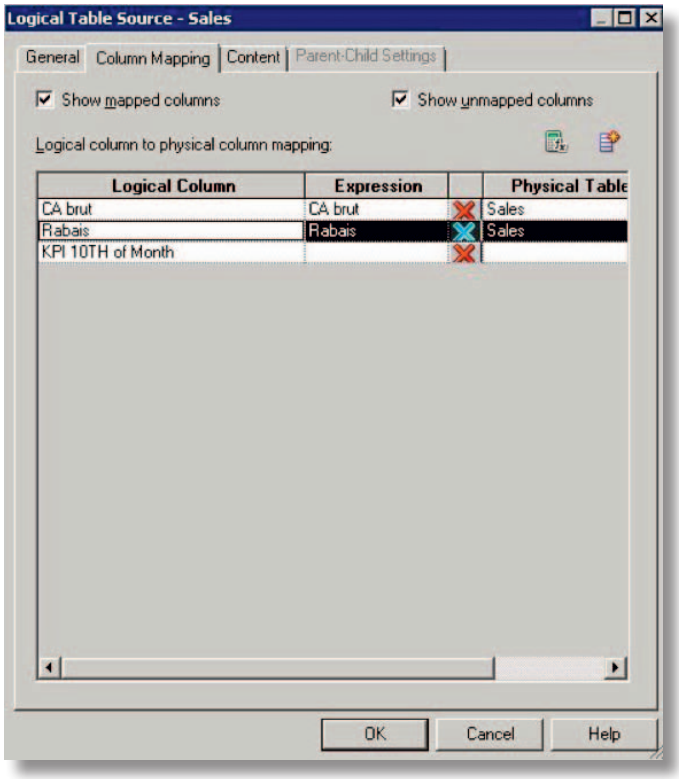
Pour chacune des sources, on retrouvera le mapping des colonnes disponibles. Celles non disponible ne seront pas renseignées.



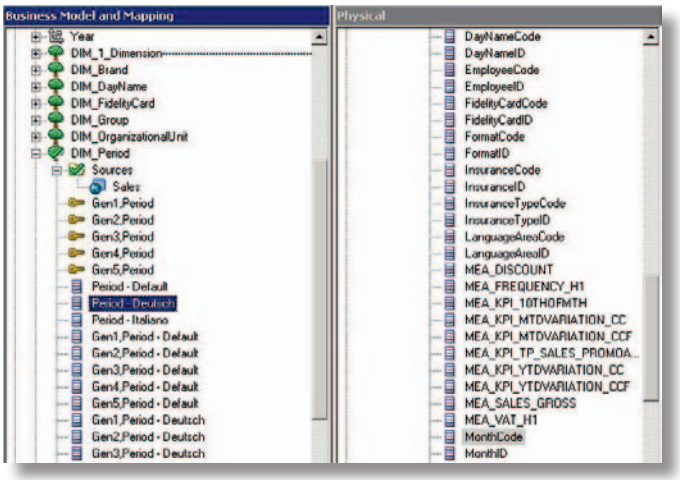


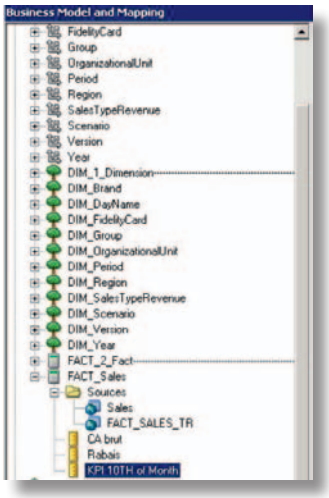
Une règle d'agrégation doit être définie sur la nouvelle colonne. Comme celle-ci est une mesure, il est important de définir «Sum» pour permettre une analyse en profondeur depuis un certain niveau. Dès cette modification faite, l'icône de la mesure change pour informer qu'une règle a été ajoutée.





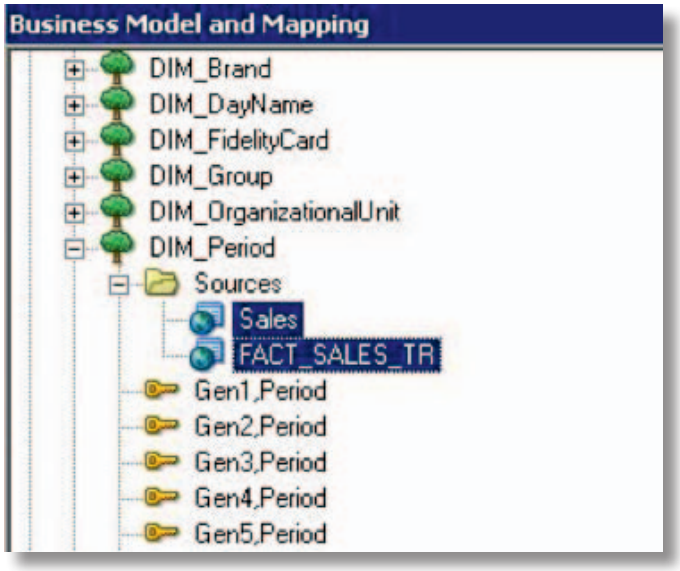
L'étape suivante est très importante pour le bon déroulement de la fédération. Il s'agit de définir les relations logiques entre la source Essbase et la source relationnelle dans la couche business.



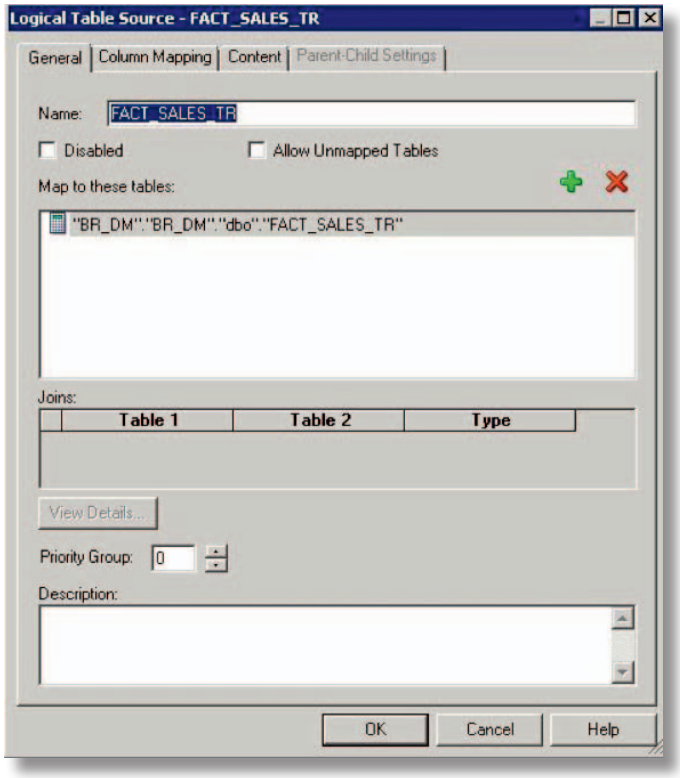


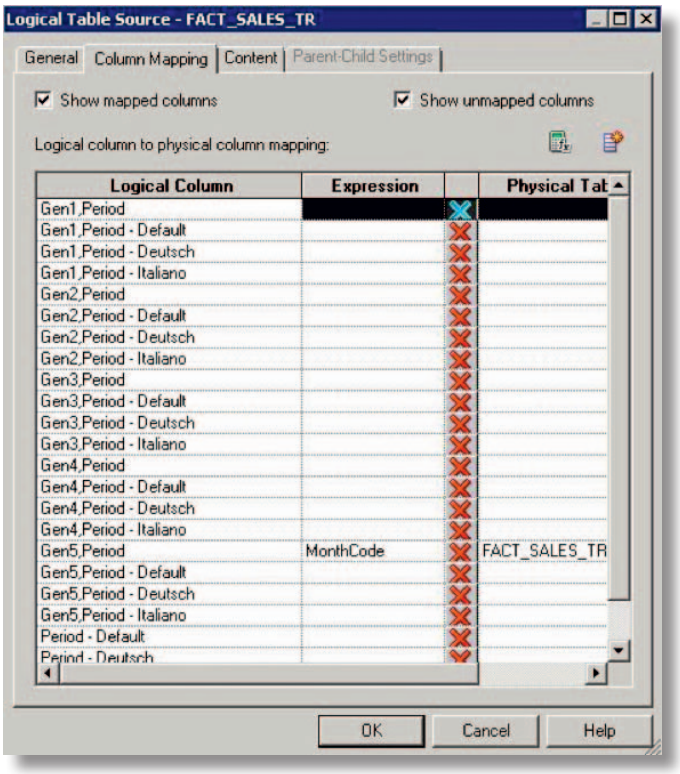
Pour ce faire, il faut glisser chaque métadonnée existante de la table des faits de notre source relationnelle sur le bon membre de la dimension correspondante dans notre couche 2.

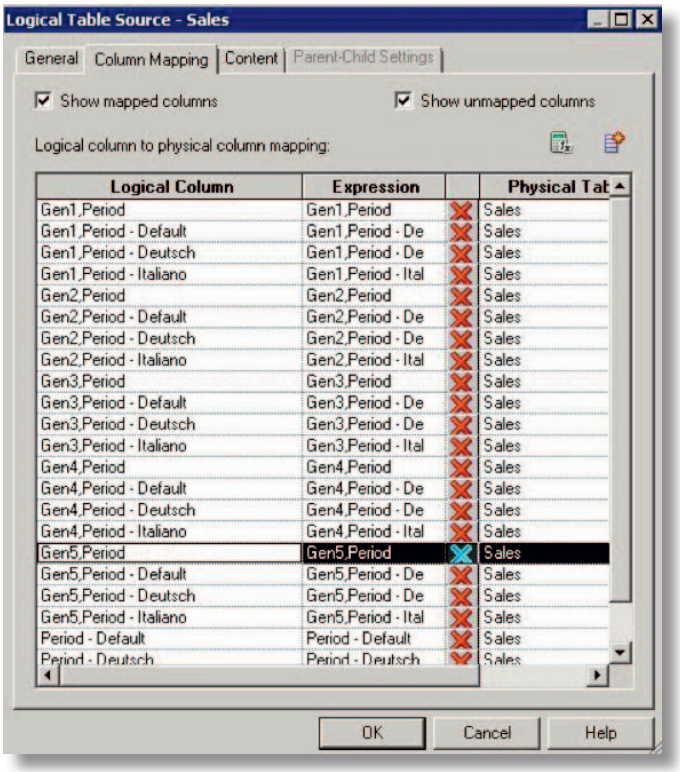
A ce moment, une nouvelle source logique se crée pour cette dimension.



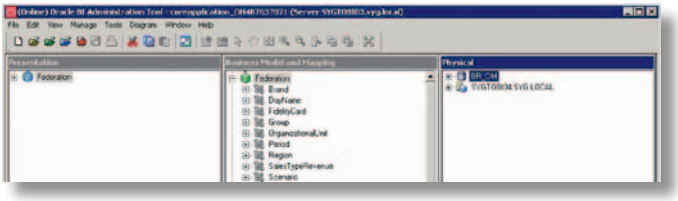
Ce répertoire «Sources» contiendra aussi les informations du lien entre la couche physique et la couche business.



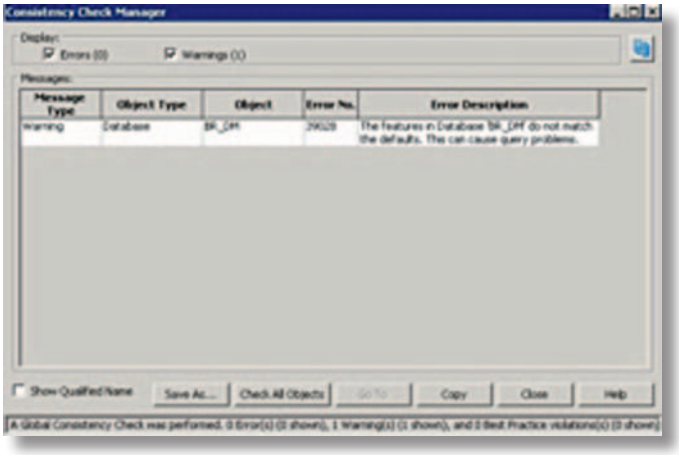




Enfin, il suffit de pousser notre business model dans la dernière couche, la couche de présentation.



Finalement, il est recommandé de lancer un contrôle global de la consistance du Repository.



Si le processus a été correctement effectué, un message informera que le business model est consistant. Sinon une fenêtre d'erreur s'affichera en mentionnant le nombre d'erreurs & de warnings.

3.4.1.3 Vérification de l'ajout de la mesure

La création d'une analyse permettant l'affichage des données provenant des 2 sources peut être défini comme suit :

Tous les champs proviennent du cube Essbase sauf la colonne «KPI 10TH of Month» qui provient de la source de données relationnelle.

federation_test

Year - Default	Gen5,Period	Gen5,OrganizationalUnit	KPI 10TH of Month	CA brut
2013	MTH_01	POS_0388AA21_067E_43A3_A2EA_77486445D250	11 180	332 706
		POS_108419EC_89C2_410F_9016_10791BA99B97	3 108	115 495
		POS_13DE29AB_39B7_40FB_BCB8_0F8CD6372BB5	8 577	267 367
		POS_155AB5B3_E64B_4929_9372_25D374915924	22 805	557 299
		POS_1695DC49_B2CF_4413_B3F4_D666ED4ABE93	11 216	377 390
		POS_16C2BE09_31D7_433E_8AF8_3B873C11CB94	11 072	157 519
		POS_1B5453E6_4E2F_498B_928B_0BECA6327C09	8 912	243 714
		POS_1CC483D7_F7F6_4E71_88CE_D572B8C5014A	4 744	179 247
		POS_27003808_AFC7_4893_AAC5_493A0737A6B6	26 852	1 062 404
		POS_330C3B54_9B05_4805_B5D4_148C6D3DC490	1 819	60 330

La requête logique envoyée par le «BI Presentation» sera la suivante :

```
SET VARIABLE QUERY_SRC_CD='Report',SAW_SRC_PATH='/users/weblogic/_filters/Federation_test/federation_test';SELECT
   0 s_0,
   "Federation"."- OrganizationalUnit"."Gen5,OrganizationalUnit" s_1,
   "Federation"."- Period"."Gen5,Period" s_2,
   "Federation"."- Year"."Year - Default" s_3,
   "Federation"."- SalesAmount"."CA brut" s_4,
   "Federation"."- SalesAmount"."KPI 10TH of Month" s_5
FROM "Federation"
WHERE
(("- Period"."Gen5,Period" = 'MTH_01') AND ("- Year"."Year - Default" = '2013'))
ORDER BY 1, 4 ASC NULLS LAST, 3 ASC NULLS LAST, 2 ASC NULLS LAST
FETCH FIRST 65001 ROWS ONLY
```

Puis le «BI Server» sera assez intelligent pour scinder en 2 requêtes la demande.

La première requête est la requête MDX envoyée au cube Essbase.

```
]]
[2014-09-29T18:24:09.000+02:00] [OracleBIServerComponent] [TRACE:2] [USER=28] [] [ecid: 4be8c04fee24b511:40a29060:1485b2f9812:-8000-0000000000e9]
Repository: Star, Subject Area: Federation, Presentation: Federation

]]
[2014-09-29T18:24:09.000+02:00] [OracleBIServerComponent] [TRACE:2] [USER=28] [] [ecid: 4be8c04fee24b511:40a29060:1485b2f9812:-8000-0000000000e9]
With
  set [_OrganizationalUnit45] as 'Generate([OrganizationalUnit].Generations(2).members, Descendants([OrganizationalUnit].currentmember, [Organiza
  set [_Period5] as '[[Period].[MTH_01]]'
  set [_Year0] as '[[Year].[2013]]'
select
  { [Measures].[MEA_SALES_GROSS]}
  ( on columns,
  NON EMPTY {crossjoin([[_OrganizationalUnit45]],crossjoin([[_Period5]],[[_Year0]]))} properties MEMBER_NAME, GEN_NUMBER, [Year].[Default] on rows
from [ES.Sales]
```

La deuxième requête est la requête SQL envoyée à la source de données relationnelle.

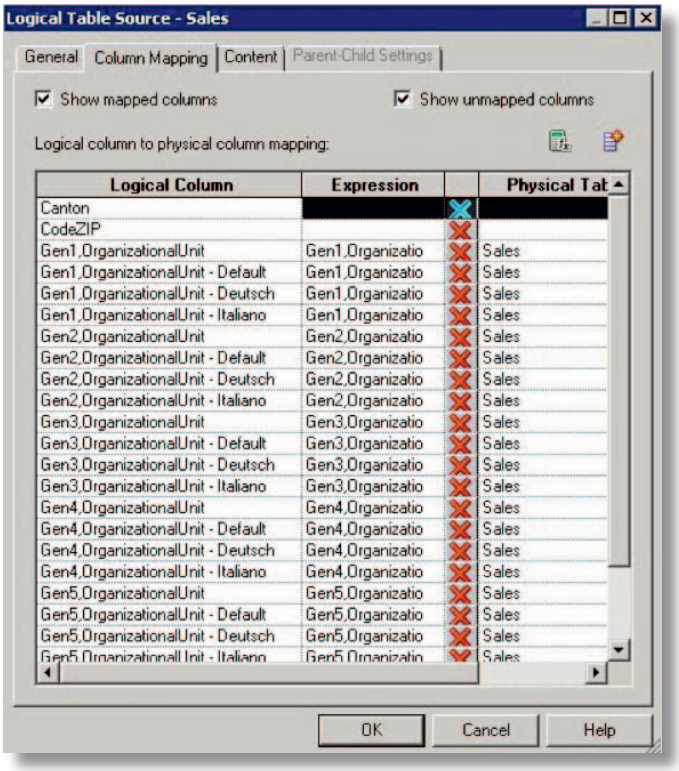
```
select sum(T30570."MEA_KPI_10THOFMTH") as c1,
     T30570."POSCode" as c2,
     T30570."MonthCode" as c3,
     T30570."YearCode" as c4
from
     "FACT_SALES_TR" T30570
where  ( T30570."YearCode" = '2013' and T30570."MonthCode" = 'MTH_01' )
group by T30570."YearCode", T30570."MonthCode", T30570."POSCode"
order by c2, c3, c4
```

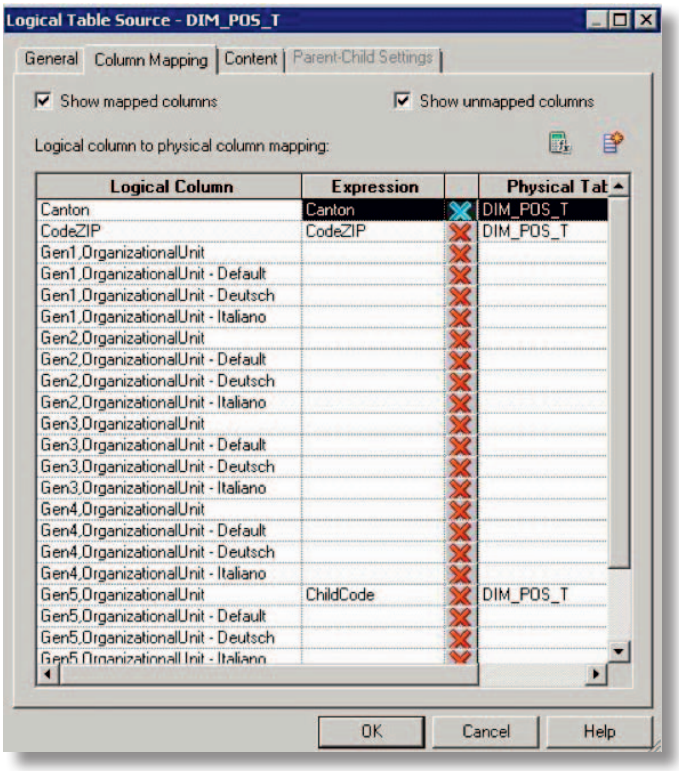
3.4.1.4 Cas pratique – Ajout d'un attribut

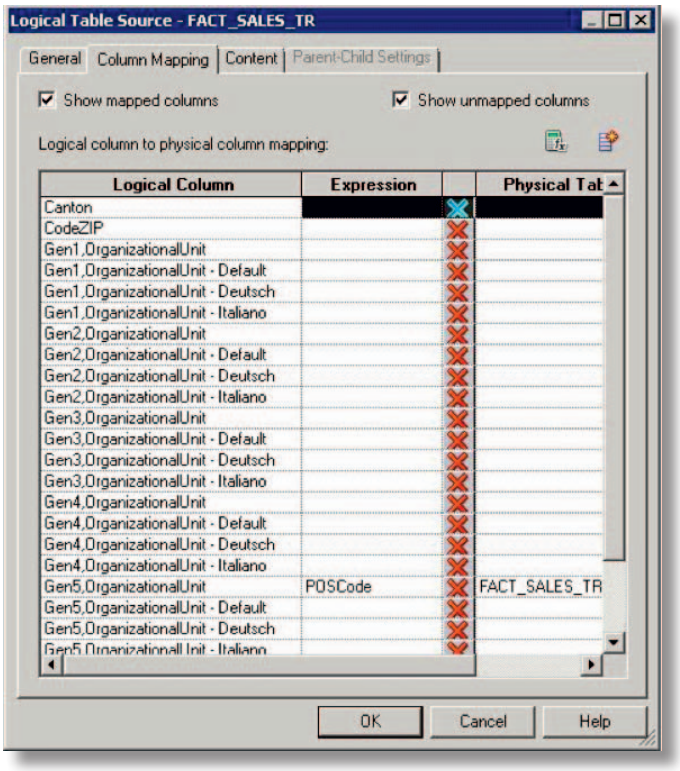
Dans cet exemple, une information supplémentaire concernant la dimension «OrzanizationalUnit» a été demandée. Nous souhaitons voir le chiffre d'affaires par point de vente et connaître son emplacement cantonal. La colonne «Canton» existe seulement dans la base de données relationnelle.

3.4.1.5 Exemple de mise en place

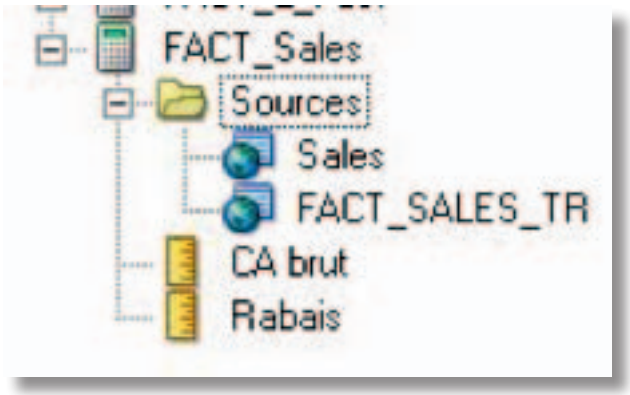
L'ajout de cette métadonnée dans le Repository est identique à ce qui a été fait dans le chapitre précédent, à ceci près que les étapes se font directement sur une dimension.

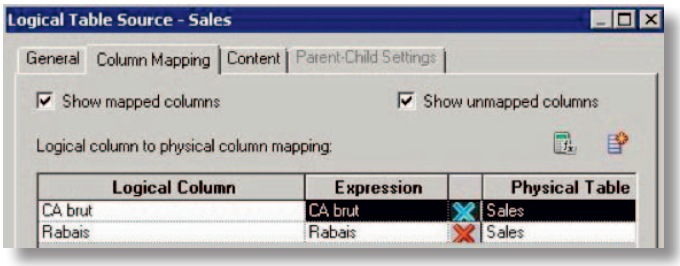


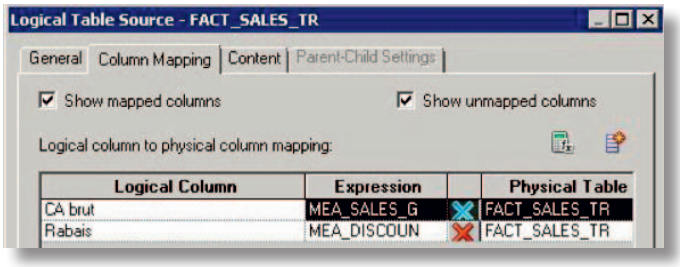




Les mesures doivent aussi être mappées sur les 2 sources pour éviter des erreurs de création de requêtes.







3.4.1.6 Vérification de l'ajout de l'attribut

La requête logique envoyée par le « BI Presentation » sera la suivante :

```
--------
SET VARIABLE QUERY_SRC_CD='Report',SAW_SRC_PATH='/users/weblogic/_filters/Federation_test/federation_Horizontal_test';SELECT
   0 s_0,
   "Federation_Horizontal"."- OrganizationalUnit"."Canton" s_1,
   "Federation_Horizontal"."- OrganizationalUnit"."Gen5,OrganizationalUnit" s_2,
   "Federation_Horizontal"."- Period"."Gen5,Period" s_3,
   "Federation_Horizontal"."- Year"."Year - Default" s_4,
   "Federation_Horizontal"."- SalesAmount"."CA brut" s_5
FROM "Federation_Horizontal"
ORDER BY 1, 2 ASC NULLS LAST, 5 ASC NULLS LAST, 4 ASC NULLS LAST, 3 ASC NULLS LAST
FETCH FIRST 65001 ROWS ONLY
```

federation_Horizontal_test

Canton	Year - Default	Gen5,Period	Gen5,OrganizationalUnit	CA brut
CTN_AG	2011	MTH_12	POS_1695DC49_B2CF_4413_B3F4_D666ED4ABE93	-1
			POS_1CC483D7_F7F6_4E71_88CE_D572B8C5014A	148
			POS_8FCB775E_809C_4682_8156_808C40204607	247
			POS_9FD80806_15FB_48BF_9004_73F6187215F0	1 110
			POS_AFD185E6_AB1D_4127_8328_086F56541250	-177
			POS_C5C7EC3F_E46A_4DF9_9B78_B60C6DBF2ABA	372
	2012	MTH_01	POS_1695DC49_B2CF_4413_B3F4_D666ED4ABE93	313 203
			POS_1CC483D7_F7F6_4E71_88CE_D572B8C5014A	151 970
			POS_8FCB775E_809C_4682_8156_808C40204607	185 736
			POS_9FD80806_15FB_48BF_9004_73F6187215F0	434 365
			POS_AFD185E6_AB1D_4127_8328_086F56541250	171 159
			POS_C5C7EC3F_E46A_4DF9_9B78_B60C6DBF2ABA	243 110
		MTH_02	POS_1695DC49_B2CF_4413_B3F4_D666ED4ABE93	304 883
			POS_1CC483D7_F7F6_4E71_88CE_D572B8C5014A	138 620
			POS_8FCB775E_809C_4682_8156_808C40204607	193 837

Modifier - Actualiser - Imprimer - Exporter - Ajouter au carnet de rapports - Copier

Puis le « BI Server » analysera la demande et il constatera qu'une information demandée se trouve dans la source de données relationnelle ; par conséquent il enverra la requête physique seulement à cette dernière.

```
]]
[2014-09-29T18:40:45.000+02:00] [OracleBIServerComponent] [TRACE:2] [USER-18] [] [ecid:
WITH
SAWITH0 AS (select sum(T30570."MEA_SALES_GROSS") as c1,
     T29303."Canton" as c2,
     T29303."ChildCode" as c3,
     T30570."MonthCode" as c4,
     T30570."YearCode" as c5
from
     "DIM_POS_T" T29303,

     "FACT_SALES_TR" T30570
where  ( T29303."ChildID" = T30570."POSID" )
group by T29303."ChildCode", T29303."Canton", T30570."YearCode", T30570."MonthCode")
select top 65001 0 as c1,
     D1.c2 as c2,
     D1.c3 as c3,
```

Si dans cet exemple, la colonne « Canton » est enlevée.

Le « BI Server », lui, interprétera les données et il enverra la requête physique seulement au cube Essbase.

3.4.2 Fédération verticale

Nous parlerons de **fédération verticale** lorsque l'on se trouvera à des niveaux différents de granularité.

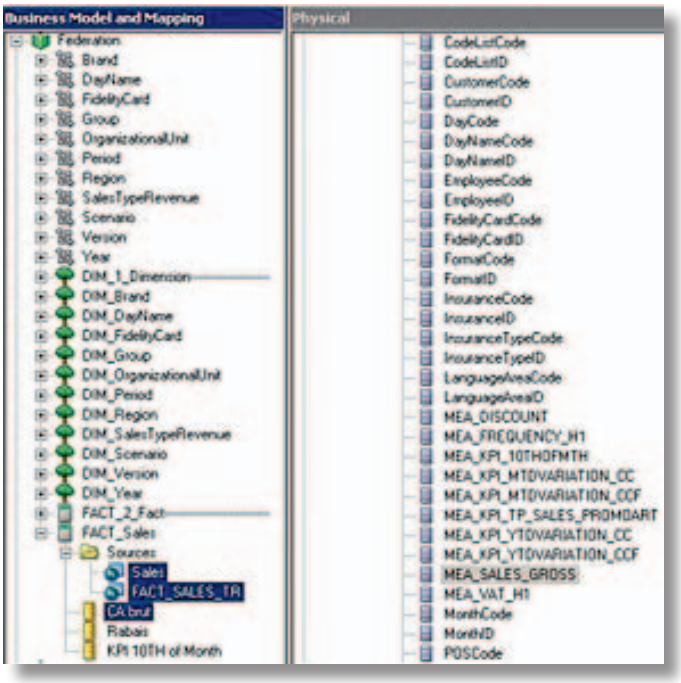
3.4.2.1 Cas pratique – Descente à un niveau atomique

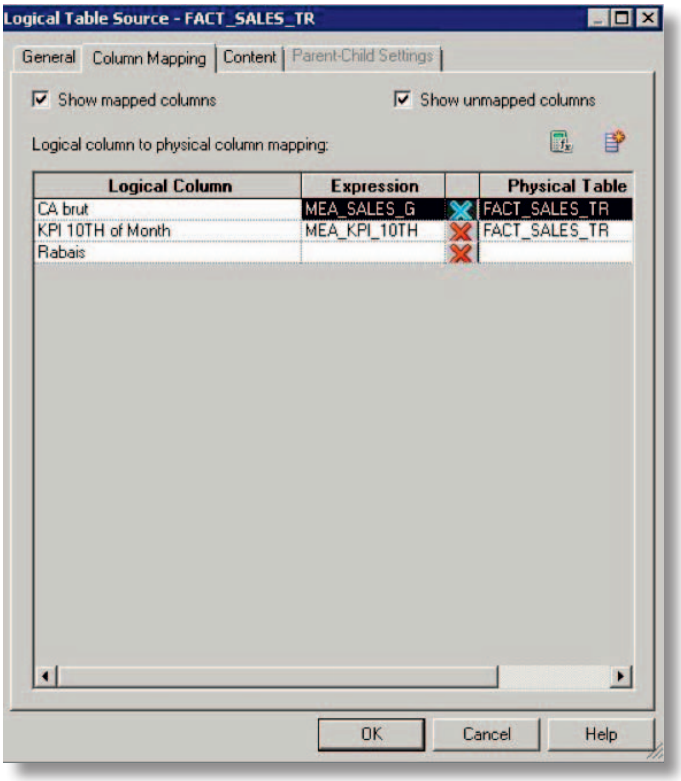
Nous avons un rapport qui décrit le chiffre d'affaires des ventes par point de vente. Maintenant nous désirons descendre à un niveau plus détaillé. Nous souhaiterons voir les articles vendus par point de vente.

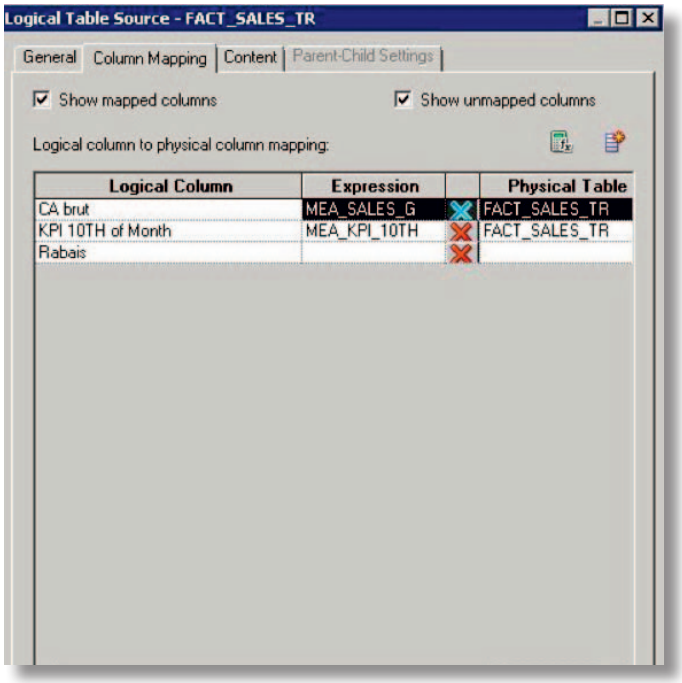
3.4.2.2 Exemple de mise en place

Le cliquer/glisser est utilisé de la même manière que lors de la fédération horizontale. Il suffit de glisser les métadonnées de la couche 1 sur la métadonnée de la couche 2.

Les premiers membres à joindre sont les mesures.

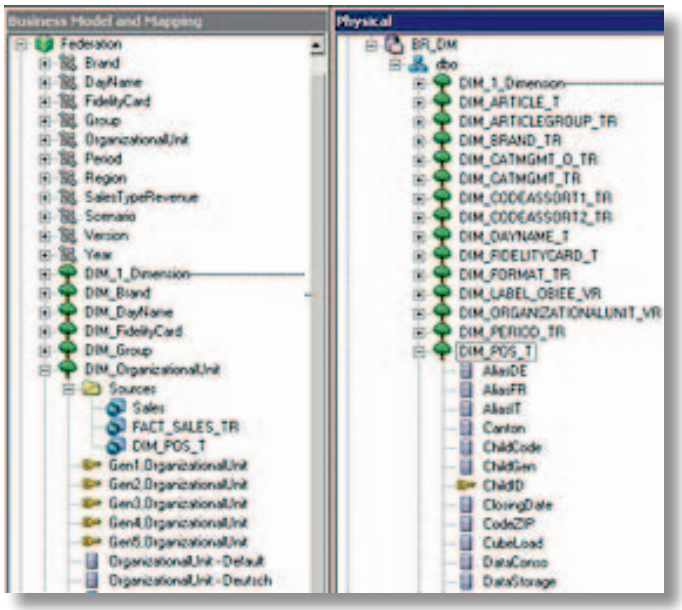






Dans le répertoire «Source» de la table de faits logique, on peut ensuite distinguer les 2 sources distinctes.

Ensuite, il sera demandé de faire des mapping sur les membres des dimensions que l'on doit utiliser pour la création du rapport. Nous allons nous occuper de la dimension «OrganizationalUnit».



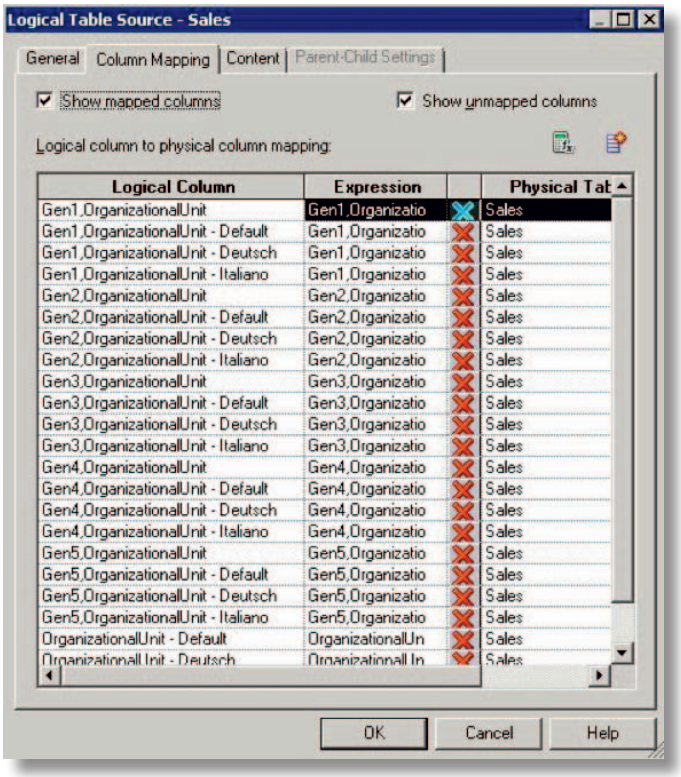
Pour la dimension, il est important d'avoir 3 jointures sur la métadonnée de la couche 2 à la couche 1.

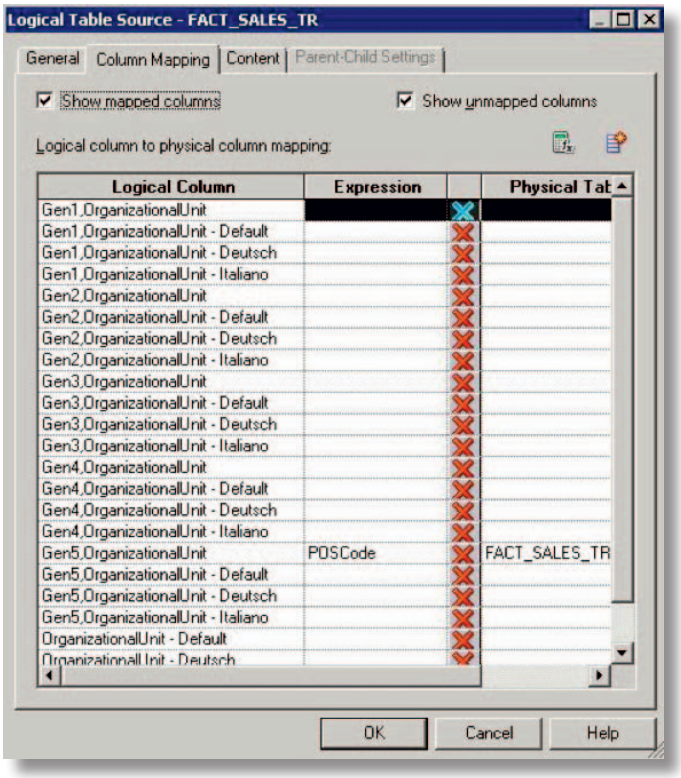
La première jointure correspond au lien à la source primaire, c'est-à-dire le cube Essbase.

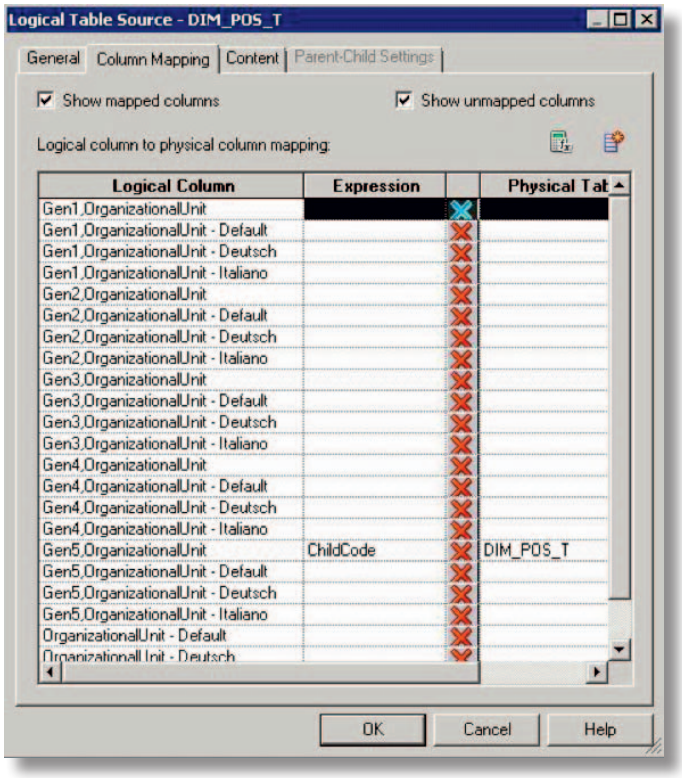
La deuxième jointure correspond au lien à la source relationnelle au niveau des faits.

La troisième jointure correspond au lien à la source relationnelle au niveau de la dimension.

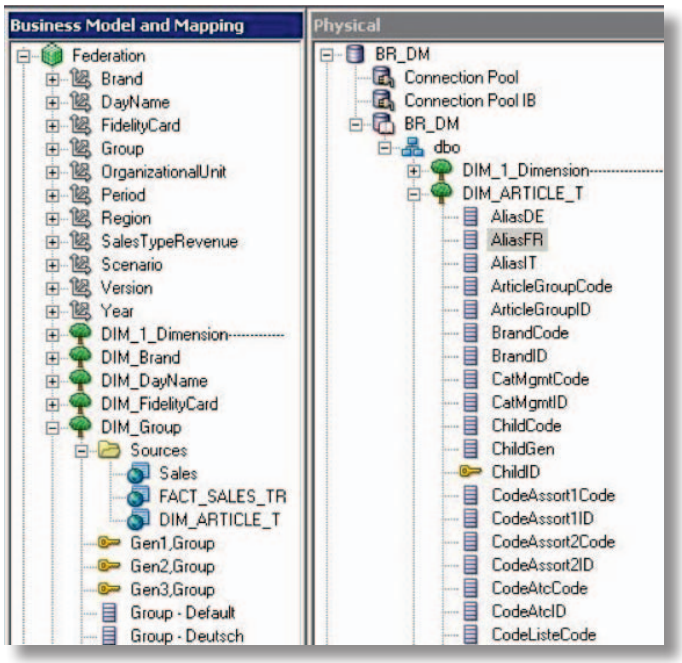
Si ces 3 liens ne sont pas faits, le BI server n'arrivera pas à traduire la demande.







Puis, nous allons procéder au même schéma pour la dimension « Group » qui contient les catégories d'articles. Cette dimension ne possède pas un niveau de détails jusqu'à l'article, sinon cela ferait beaucoup trop de membres à stocker dans le cube. Surtout que le nombre d'articles s'accroît toujours.



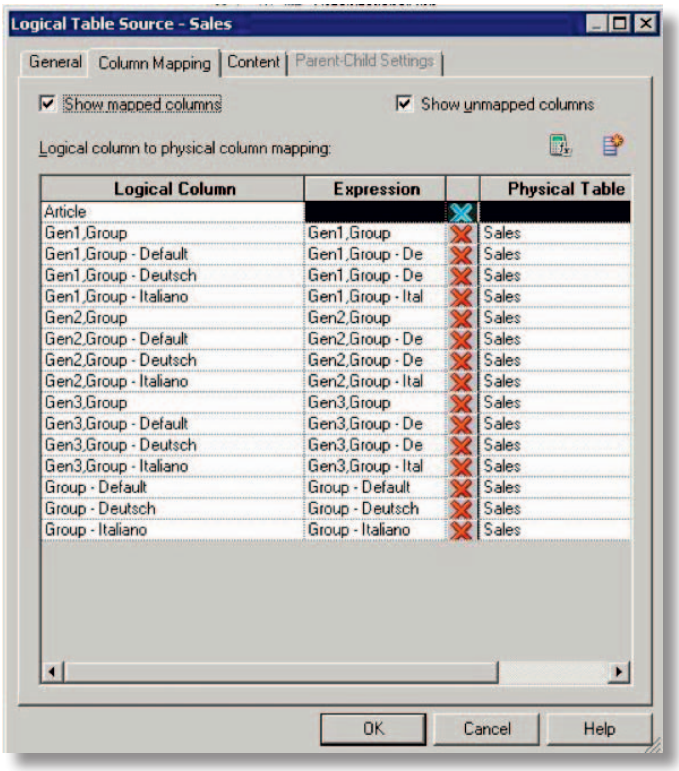
Dans ce cas, nous avons aussi 3 jointures.

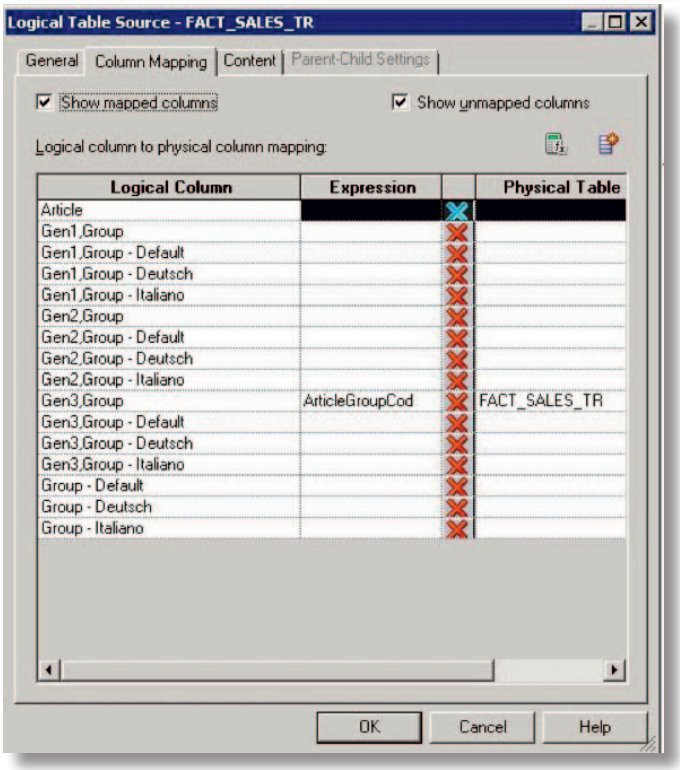
La première jointure correspond au lien à la source primaire, c'est-à-dire le cube Essbase.

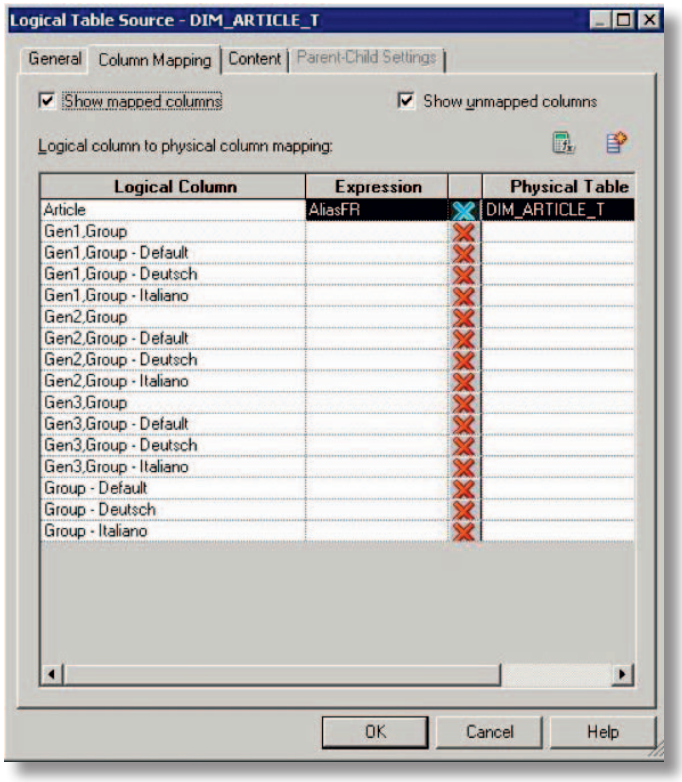
La deuxième jointure correspond au lien à la source relationnelle avec le niveau le plus bas de la catégorisation provenant de la table des faits

La troisième jointure correspond au lien à la source relationnelle pour aller chercher l'article.

Si ces 3 liens ne sont pas faits, le BI server n'arrivera pas à traduire la demande.

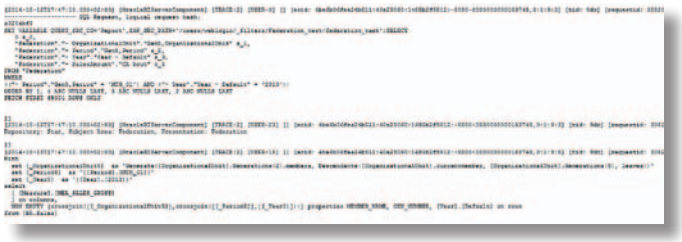


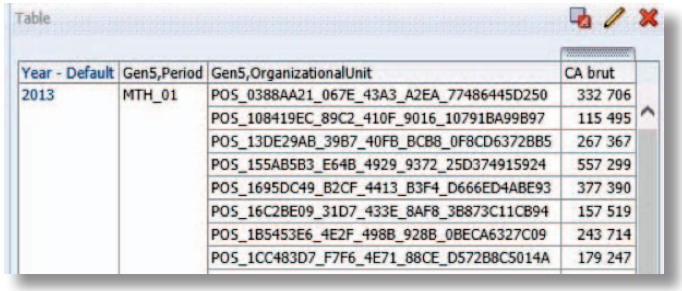




3.4.2.3 Vérification de la descente à un niveau atomique

Dans une première vue, nous allons afficher le rapport sans article.



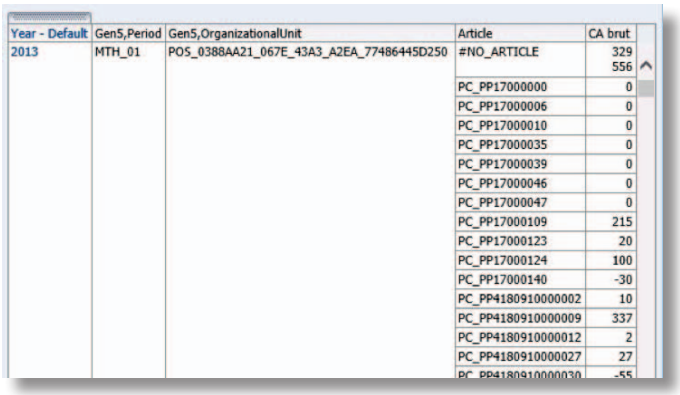


Table

Year - Default	Gen5,Period	Gen5,OrganizationalUnit	CA brut
2013	MTH_01	POS_0388AA21_067E_43A3_A2EA_77486445D250	332 706
		POS_108419EC_89C2_410F_9016_10791BA99B97	115 495
		POS_13DE29AB_39B7_40FB_BCB8_0F8CD6372BB5	267 367
		POS_155AB5B3_E64B_4929_9372_25D374915924	557 299
		POS_1695DC49_B2CF_4413_B3F4_D666ED4ABE93	377 390
		POS_16C2BE09_31D7_433E_8AF8_3B873C11CB94	157 519
		POS_1B5453E6_4E2F_498B_928B_0BECA6327C09	243 714
		POS_1CC483D7_F7F6_4E71_88CE_D572B8C5014A	179 247

Nous pouvons constater que le BI server est assez intelligent pour envoyer une requête MDX au cube Essbase seulement.

Maintenant, si je désire ajouter l'article.



Year - Default	Gen5,Period	Gen5,OrganizationalUnit	Article	CA brut
2013	MTH_01	POS_0388AA21_067E_43A3_A2EA_77486445D250	#NO_ARTICLE	329 556
			PC_PP17000000	0
			PC_PP17000006	0
			PC_PP17000010	0
			PC_PP17000035	0
			PC_PP17000039	0
			PC_PP17000046	0
			PC_PP17000047	0
			PC_PP17000109	215
			PC_PP17000123	20
			PC_PP17000124	100
			PC_PP17000140	-30
			PC_PP4180910000002	10
			PC_PP4180910000009	337
			PC_PP4180910000012	2
			PC_PP4180910000027	27
			PC_PP4180910000030	-55

Le BI server comprend qu'une colonne doit être recherchée dans le DataMart et il envoie une requête SQL.

Conclusion

Les deux concepts présentés dans cet article démontrent qu'il est possible de faciliter la vie des utilisateurs finaux en leur fournissant un accès à différents jeux de données (agrégées ou détaillées) au travers d'un seul composant de restitution : SmartView ou OBIEE. La quantité de données gérées par ces modules nécessite parfois l'utilisation de nombreuses ressources. Pour répondre à cette problématique, Oracle a certifié la suite OBIF sur son Appliance Exalytics. Les concepts présentés dans cet article se combinent sans problème avec la technologie Exalytics. ■

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